FIFA

HAWK-EYE
INNOVATIONS

TECHNICAL & PRODUCTION GUIDE V1

FIFA WORLD CUP 2026

11.06.26 - 19.07.26

PRIVATE & CONFIDENTIAL

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During this tournament, we shall be using the official name of the stadium and not the sponsorship name. The official name shall be used in smart files, documentation including match reports etc

VENUE	Stadium Name	Official Name for FWC26	REGION	CODE	VENUE NUMBER	
IBC	Kay Bailey Hutchison Convention Center			IBC	0	N/A
NEW YORK NEW JERSEY	MetLife Stadium	**New York New Jersey Stadium**	EASTERN	NYNJ	1	
MIAMI	Hard Rock Stadium	**Miami Stadium**	EASTERN	MIA	2	
ATLANTA	Mercedes-Benz Stadium	**Atlanta Stadium**	EASTERN	ATL	3	
DALLAS	AT&T Stadium	**Dallas Stadium**	CENTRAL	DAL	4	
KANSAS CITY	Arrowhead Stadium	**Kansas City Stadium**	CENTRAL	KC	5	
LOS ANGELES	SoFi Stadium	**Los Angeles Stadium**	WESTERN	LA	6	
BOSTON	Gillette Stadium	**Boston Stadium**	EASTERN	BOS	7	
VANCOUVER	BC Place	**BC Place Vancouver**	WESTERN	VAN	8	
SEATTLE	Lumen Field	**Seattle Stadium**	WESTERN	SEA	9	
MEXICO CITY	Estadio Azteca	**Mexico City Stadium**	CENTRAL	CDMX	10	
PHILADELPHIA	Lincoln Financial Field	**Philadelphia Stadium**	EASTERN	PHL	11	
HOUSTON	NRG Stadium	**Houston Stadium**	CENTRAL	HOU	12	
TORONTO	BMO Field	**Toronto Stadium**	EASTERN	TOR	13	
SAN FRANCISCO	Levi's Stadium	**San Francisco Bay Area Stadium**	WESTERN	SFBA	14	
MONTERREY	Estadio BBVA	**Monterrey Stadium**	CENTRAL	MTY	15	
GUADALAJARA	Estadio Akron	**Guadalajara Stadium**	CENTRAL	GDL	16	

INTRODUCTION

Introduction

This guide should be read by all staff operating at the **FIFA World Cup 2026 USA, Canada & Mexico (FWC26)**. It contains important information on the technical setup of our operational delivery across all services.

This guide should be read in conjunction with the tournament specific Centralised Sheet (for logistics, travel, accommodation and H&S information). An electronic version of this guide can be found on the project folder on the Google Drive which includes link to other documents as necessary. The electronic version supersedes any printed versions and will show all relevant updates.

If you cannot access the electronic copy from the link above, then please inform the FWC26 Management team (FWC26.Management@hawkeyeinnovations.com).

This guide, and any links to documents within it, should provide you with all the information that you need to know. There will be no separate technical guides or setup documentation produced. This guide covers both onsite and IBC operations

Please keep this guide secure at all times given the sensitive information that it contains.

Important Change to the System - Please Read!

This World Cup will utilise **centralised processing** (GLT & EPTS/AOT) and **centralised VAR capture**.

Data & Feed Workflow

Tracking camera feeds from the venue are streamed across the HBS-provided data link. VAR feeds are provided to us in the IBC (only 6x backup feeds onsite). RRA output shall be streamed from the IBC to the venue over the data link. TV feeds are provided across both locations; GLT L and R Box cams will be provided to HBS at the venue. VAR Replay and GLT/AOT VR Replay feeds will be provided at the IBC.

Hardware:

- **1x rack per venue (3x servers)**

  The servers will be running ZCam mirror app as well as capturing 6x backup video feeds and running a backup viewer
- **6x "match kits" at the IBC:** Covering GLT, EPTS, VAR.
- **SMARTViewer:** for 7x replay devices shall run on machines based in the IBC. The remote users (venues) will use SMARTRemote to access the viewers over the link

Match Designations (A-F)

For each day, matches scheduled for that day will be given a match designation (A-F). Unless under DRP, these correspond to the VOR allocations; i.e., Match A will be in VOR 1, Match B in VOR 2, etc. These also correspond to the circuits on which the video feeds from HBS will be provided (at the IBC) and are in chronological order by match KO times (Dallas time). They will also be noted on the match checklist.

For the majority of days, there is a maximum of 4x matches taking place. Matches E and F shall be used on the final group stage days. In some cases, Matches A and E are grouped and Matches B and F are grouped (e.g. the 12G UHD PGM feed for the VORs is only available as A/E, B/F, C, D distribution.

SERVICE OVERVIEW

VAR

- Centralised VAR Processing & Operations NEW
- Kinexon Integration
- GK Encroachment
- Ball Out of Bounds Detection NEW
- GK Line of Sight Feed NEW
- MVAR & SIMVAR Training

TRACKING

- Centralised GLT & EPTS Processing & Operations NEW
- Goal Line Technology with Leikr watches
- Enhanced Semi Automatic Offside Officiating (AOT)

BROADCAST

- Combined VR (GLT and ASAO VR in 1 System) NEW
- Automatic Offside VR Replays NEW
- ASAO VR Replays over 16m Cameras NEW
- 3D Player Scans with Player Avatars used in VR Replays NEW

DATA

- Event Data Services
- QC/Scrubbed Player Data

OTHER

- Analyst/Medical Replay Devices
- Referee Camera Viewers (IBC)
- Centralised Injury Spotter (IBC)
- Fast Feed Data for HBS (dynamic audio mixing for ball position) NEW
- Generation of CIS and Subs Feeds
- Feeds for FOC (FIFA Operations Centre) at IBC
- LiveAR for Multilateral Production - 6x Distributions
- LiveAR for Unilateral Production, DAZN Japan- 104x Matches NEW
- Cloud Video Replay (proof of concept) NEW

IBC: International Broadcast Centre

The International Broadcast Centre (IBC) is situated in the Kay Bailey Hutchison Convention Centre in Dallas, Texas.

The IBC will serve as the central hub for HBS, broadcast rights holders, and media. For this World Cup, HEI has significantly centralised its operations.

Hawk-Eye's presence is split across two locations:

Zone D:
The FTI (Football Technology Innovations) area is located here.

HBS Digital Area:
The HBS Datatainment room is within this area.

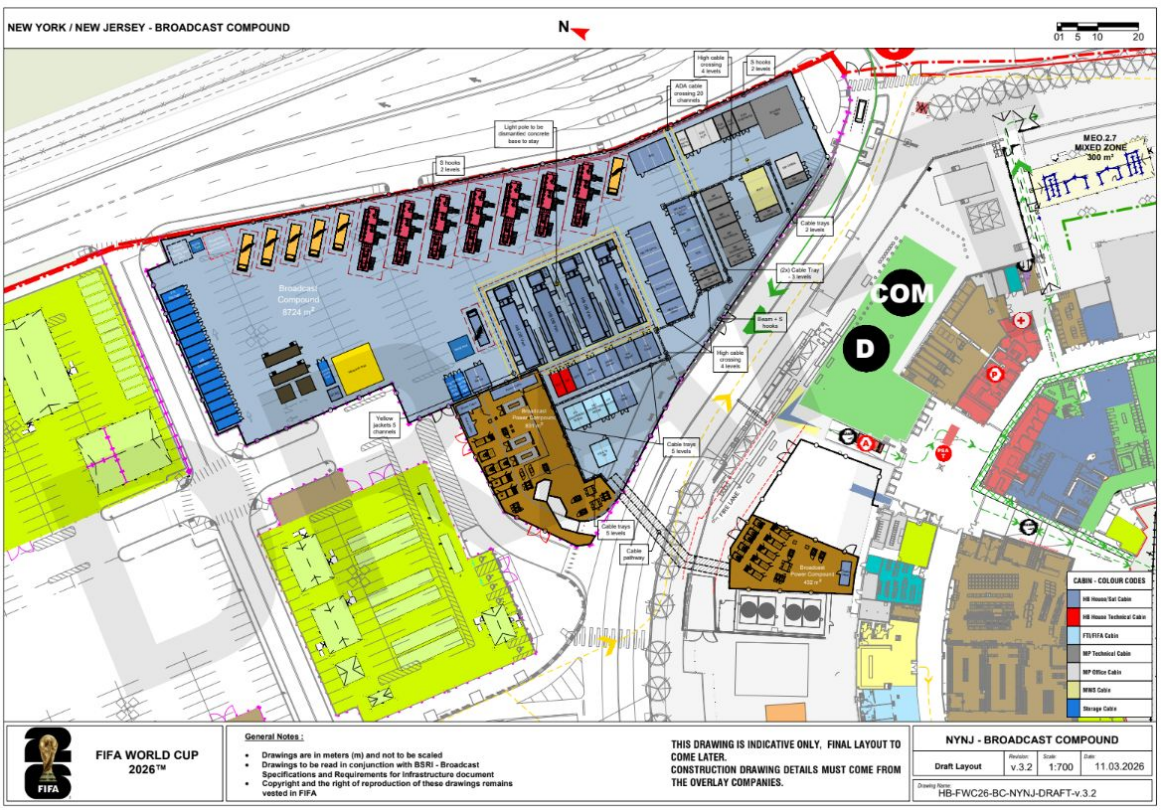
Venue OB Compounds:

Plans:
Key:
FTI = FIFA Technology & Innovation [team]
CER = Central Equipment Room
TOC = Technical Operations Centre
TER = Technical Equipment Room
TIR/CIR = Interconnect Rooms
MRL = Media Rights Licensee (i.e. a broadcaster with rights of coverage)

FTI Fibre [OB to Catwalk, Media Tribune & Pitchside cabinets] Connections:

Fibre connections will be terminated into the FTI Equipment Room in the OB Compound. The fibre installation is not managed by HEI. The installation will be made by FIFA's chosen provider and ready no later than MD-7 at the required locations. An overview of the fibre locations & fibre core plug up per venue can be found on the "Fibre" tab at the FWC26 - Stadium Installation Overview document.

OB Compound Plans can be accessed HERE.



Hawk-Eye Team - Project Management

- **Project Manager:** **Nathan Bedford**

- **Technical Manager:** **Harry Sainsbury**
- **Installation Manager:** **Lee Jones**
- **Tracking Operations:** **Dom Radford**
- **Data Operations:** **Sam Rance**
- **RO Lead**: **Samir Baza**
- **Events Team Management**: **Laurence Upshon**
- **Events Team Management**: **Matt Skevington**
- **Logistics Manager:** **Louise Lynn**

Hawk-Eye Team - Venue Based

The team structure for FWC26 has been implemented based upon feedback and learnings from previous FIFA tournaments as well as UEFA matches and similar tournament deliveries. Note that there is a reduced staffing model onsite to take into account centralised processing / VAR capture.

Venue Manager:
The venue manager has overall responsibility for **all** operations and service delivery at the venue. They will be the main point of contact for FIFA and HBS when onsite and will monitor all onsite deliverables. They are responsible for ensuring a smooth delivery of service. The VM is responsible for general video services such as setting up replay devices, RRA/BURRA, setup/checks and handover/collection of the relevant video feeds onsite. The VM is also responsible for the backup VAR station, H&S, scheduling, general welfare of the venue team and final escalation.

Tracking Technical Guarantee:
The Tracking Technical Guarantee (TTG) will assist the IBC GLT Operator & IBC EPTS/AOT Operator during the setup and operation of the systems. The TTG is responsible for overseeing all the tracking systems hardware onsite, conducting the on-pitch calibration and pre-match referee (GLT) test / handover of watches etc. They can also assist the VM in the setting up of pitchside equipment such as replay devices and the RRA/BRRA etc.

Event Data Collector:
Responsible for collecting match event data using the FELA application. The EDC will be in contact with the Event Data Reviewer (EDR) located in the IBC. They will also be given duties by the VM to support the wider delivery of services.

Hawk-Eye Team - IBC Based

Event Management, Duty Managers, Technical Managers:
Senior management, final point of escalation and responsible for overall delivery. People from each group will be assigned each day to cover Day/Night shifts along with a clear point of escalation plan.

Quality Control Manager:
Responsible for the consistency and quality across all our outputs (GLT, VAR, SAO, EPTS, Data), both technically and operationally, and adherence to HEI standards and FIFA guidelines. The QCMs play a key part of ensuring we maintain our high standards and will work with the IBC team and and Venue Managers to feedback common themes or issues based on observations. 2x QCMs (one dedicated to officiating technologies, the other dedicated to data delivery) will be located in the Football Operations Centre during live operations and will be the primary point of contact for FIFA for any issues, feedback or updates during the match periods. They will also be responsible for monitoring checklists, logging issues and match reporting.

VOR TGs:
Responsible for the technical setup, configuration and operation of the VAR system for a match. The VTG's will work on assigned matches on a rota, working in the TechOps room located in the IBC. They will be responsible for liaising with the onsite team, ensuring the VAR setup for that match is complete and correct. They will also work closely with the EPTS Supervisor for AOT integration.

Replay Operator (RO):
The main replay operator providing the VARs the tools they need to make VAR decisions. Likely to arrive between 3-4 hours before K/O and will be responsible for final checks, DRP testing and post-match clips/log completion.

Assistant Replay Operator (ARO):
Responsible for setting up the VAR system in the build-up to the match by working alongside the VOR TG to complete sync, pans, internal testing, checklists etc. Assisting the RO during live operations and relays any issues to the ARO 2/TG

Assistant Replay Operator 2 (ARO2/ TG):
The ARO2 will join approx 3-4 hours before K/O to perform QA checks. During match operations, the ARO2 will monitor and operate the Kinexon Viewer for ball telemetry data. They will also be the eyes and ears of the VOR TG and will be the point of escalation for any technical issues. The ARO2/TG may be assigned up to 2 matchers per day.

GLT Operator:
Responsible for all aspects of the GLT system. Generally, the GLT Operator will also be heavily involved with the setup calibration and FIT of the GLT system by working closely with the onsite TTG. Also responsible for colour balancing the goal line cameras.

AOT Operator:
Responsible for all aspects of the AOT system at the venue. Generally, the AOT Operator will also be heavily involved with the setup and calibration of the system by working closely with the onsite TTG. They will be responsible for monitoring the system during the match and reporting post match. Also responsible for colour balancing the cameras.

VR Operator:
Responsible for operating the combined GLT/AOT VR system. They will work closely with FIFA FQCM and broadcast to ensure replays are generated and sent quickly and accurately for all applicable events. The VR Operators will work in the VOR at the IBC and will be assigned between 1 and 2 matches per day.

Hawk-Eye Team - IBC Based

Tracking Supervisor:
Supervise GLT, EPTS and (S)AOT tracking systems, including setup, calibration, approvals, FITs, match operations and level 1 data review. Tracking Supervisors are assigned on a match by match/shift basis, rather than being assigned a venue for the duration of the tournament.

Data Supervisors:
The data supervisors are responsible for monitoring our data output to consumers, and integration with various stakeholders across player and event data. They will be monitoring the various data rates, consumers, queues and technical elements of the data handover. They are assigned on a match by match/shift basis, rather than being assigned a venue for the duration of the tournament.

Event Data Reviewers:
The EDRs will be based in the Data Operations room in the IBC and work closely with the Event Data Collectors onsite to produce accurate and timely event data statistics for the match.

Event Data Speakers:
The Event Data Speakers will be based in the Data Operations room in the IBC and work closely with the Event Data team (onsite and within the IBC) to produce accurate and timely event data statistics for the match.

Scrubbers:
The EPTS scrubbers will be based in the Data Operations room and will work with the EPTS operators to ensure accurate and timely QC'd data is delivered to FIFA from the EPTS system, in accordance with the Service Level Agreement.

Datatainment Operators:
The datatainment operators are based in the HBS area and shall monitor and operate the system in order to produce the relevant multilateral and unilateral LiveAR feeds for the Media Rights Licensees.

Logistics Managers:
There will be 3 Football Logistics Managers at the start of the event to provide support for logistical issues including freighting, accomodation and travel. Louise will be based in IBC to cover those working at the IBC. David will be based in Mexico City covering 8 venues and Dan will be based in Philadelphia covering another 8 venues.

Football Tech Managers:
The Football Tech Managers (GLT, EPTS and Data) are located at the IBC to ensure a high quality use of the various elements of technology, especially new deliverables.

Support Staff / Developers / Product Owners / Delivery Managers

Given all of the new deliverables and the importance of the FIFA World Cup, we have a wide range of support based in multiple locations to create a 24/7 support system for the opening 2 weeks of the event.

Representatives from Tracking, SMART and Data squads will ensure a high quality rollout of new deliverables. They are on hand to support these rollouts as well as to implement critical fixes or updates as required. Delivery Managers will help coordinate the delivery of new software and features within this project.

Dallas
We will have all-round support, product owners and delivery managers based in the IBC at Dallas.

Atlanta
We will have developers based in the Atlanta office to allow them to work within the same time zone

Basingstoke
We will have developers based in Basingstoke to allow work to continue to work 24/7

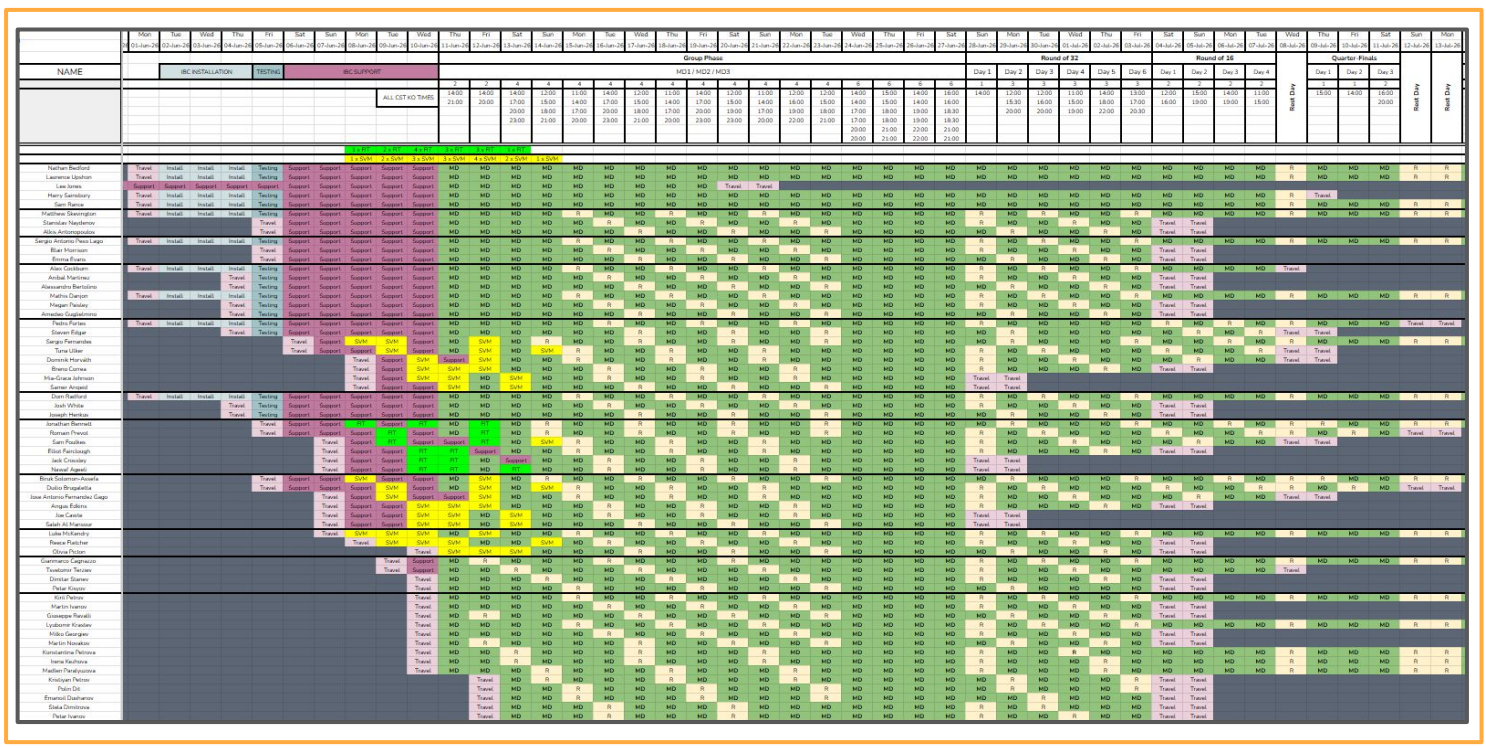
Day by Day Schedule

The provisional day by day schedule will be initially be published by the Project Manager and will be available from a link on the Centralised Sheet.

IBC:
Schedules for those based at the IBC shall be managed by the PM / Duty Manager / QCMs. Generally, call times will be published by 6pm the day before. Days off shall be known several days prior.

Venue Teams:
Once onsite, the Venue Manager will be responsible for the scheduling of the venue team. The Venue Manager will ensure that adequate rest days are provisioned for the team as based on the day by day schedule. The VMs will also be responsible for issuing call times for the onsite team and ensuring the day by day schedule is updated where necessary.

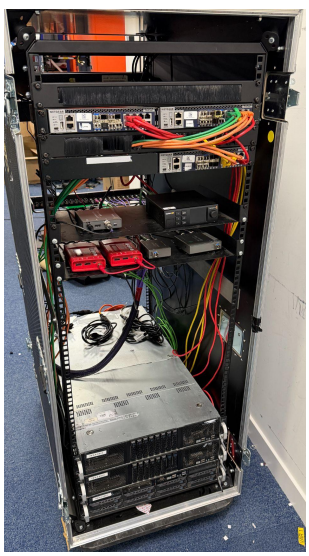
Replay Operators:
Replay Operator schedules including match assignments will be dictated by FIFA Refereeing in coordination with the RO Lead. These will also be overseen by the PM.

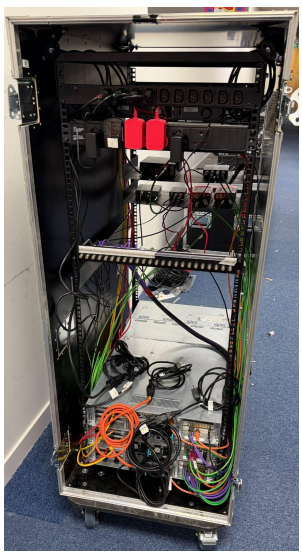


Equipment Freighting & Storage

Identification:
Colour coding shall be used on flight-cases, racks and pallets to identify the venue. The VM must ensure that these colour codes are correct prior to pickup

Venue	Stadium	HBS Code	Venue Number	
IBC	Kay Bailey Hutchison Convention Center	IBC	0	N/A
New York New Jersey	New York New Jersey Stadium	NYNJ	1	
Miami	Miami Stadium	MIA	2	
Atlanta	Atlanta Stadium	ATL	3	
Dallas	Dallas Stadium	DAL	4	
Kansas City	Kansas City Stadium	KC	5	
Los Angeles	Los Angeles Stadium	LA	6	
Boston	Boston Stadium	BOS	7	
Vancouver	BC Place Vancouver	VAN	8	
Seattle	Seattle Stadium	SEA	9	
Mexico City	Mexico City Stadium	CDMX	10	
Philadelphia	Philadelphia Stadium	PHL	11	
Houston	Houston Stadium	HOU	12	
Toronto	Toronto Stadium	TOR	13	
San Francisco	San Francisco Bay Area Stadium	SFBA	14	
Monterrey	Monterrey Stadium	MTY	15	
Guadalajara	Guadalajara Stadium	GDL	16	





Packing pictures can be found here

Equipment Freighting & Storage

Freighting Pre Event:

The equipment will be arriving across both sea and air freight

- IBC Equipment has already arrived and built during a Pre-Rig period in May
- Venue kits (cameras, install and venue base end), and MVAR- ETA early May (depending on installation and setup dates)

Racks have been configured and QAd in Basingstoke ahead of shipment.

Equipment Delivery & Post Event:

The Project team will coordinate all deliveries and collections onsite with venues. There is no expectation on VMs or Installation Staff to arrange this.

Freighting Post Event:

Venue equipment post event will leave via air freight. All IBC equipment shall leave via sea freight. The exception to this is the IBC kit that arrived from Mexico Play-Off Tournament; this will be sent back separately via air due to import/export procedures. Further plans will be communicated about this ahead of the derigs in case venue teams need to prepare equipment accordingly.

Storage:

Please keep any flight-cases, pallets (that equipment came on), rack ends, pallet cages, boxes, etc neat and tidy at all times. It is likely these can all be stored in the FTI Storage Container in the TV compound; the VM should liaise with the FTI team onsite.

It's highly recommended that you take photos of the equipment when you receive it, and use these to repack the kit after the event. Due to customs processes, kit should be packed and returned as it was received.

Venue Manager Responsibilities:

- Remove / wipe old labels from the flight cases
- Take photos of the cases / pallets prior to pickup
- Ensure all cases / pallets etc are adequately labelled (this will be communicated during the latter stages of the tournaments);
- Group the kit together after the event if no-one from HEI will be onsite for pickup. Ensure your stadium contact is aware.

Referee Training Seminar: 1st June - 7th June

Hawk-Eye has been commissioned to provide referee training seminar services pre-tournament only. This will take place in Miami.

The complete RO team have been assigned. Samir the lead RO will be leading the team to setup and deliver as well as undergo internal training to ensure everyone is well equipped and prepared ahead of the FIFA World Cup.

The MVAR kit has undergone comprehensive maintenance prior to this event and will include spare servers.

MVAR Training

- 2x MVAR system
  - From 04.06.2026 - 12.06.2023
    - Setup 01.06.2026 - 03.06.2026

- Setup (per MVAR)
  - 10x IP cameras on tripods
    - 9x static
    - 1x remotely controlled
  - 3-5x SDI camera feeds (provided by FIFA and handed over to HEI at the MVAR capture rack)
  - VAR and AVAR station, with RO and ARO
  - Same workflow as FWWC23
- 1x RRA screen on desk (pitch side)
- Integration of Kinexon system (with ARO accessing telemetry data)

Audio / ref comms will be provided Riedel.

Simulator

- 1x VAR simulator (tournament setup: VAR, AVAR, RO, ARO & ARO2) - Full setup.
- Library of clips have been prepared by Samir and are packed into one of the MVAR kits.

MVAR / Simulator

2x MVAR systems shall be setup at Miami Dade College Kendall Campus (above).

The simulator shall be setup at the FIFA Refereeing Hotel, Grand Beach Hotel, Miami Beach.

The general programme can be found here.

Pitch layout and camera plan can be found here.

Key Links

IP List - Can be found here

Venue Technical Planning - Can be found here

IBC Technical Planning incl routing template - Can be found here

Match Report - Can be found here - To be added soon

Checklist - Can be found here - To be added soon

DRP - Can be found here - To be added soon

VAR Archiving Guide - Can be found here

HBS OB Compound Layouts - Can be found here

HBS Camera Plan - Can be found here

Stadium Infrastructure and Install plans - Can be found here

Centralised Sheet - Can be found here

All links will be available on the FWC26 centralised sheet once available

TOURNAMENT KICKS OFF IN 28 DAYS!

SCHEDULES & LOGISTICS (Flights, Accommodation, Staffing Plans, Freighting, Pre Event Checklist)		OPERATIONAL DOCUMENTS (Technical Guide, Installation Reports, Checklists, DRPs, Kit Lists, H&S)	
Pre-Event Checklist	~~Available~~	Stadium Installation Guides	~~Available~~
IBC Shift Times	~~Available~~	Technical & Production Guide	~~Available~~
Event Schedule	Available	Event Master Checklist	~~Available~~
Flights & VISAs	Available	Disaster Recovery Plans	~~Available~~
Accommodation & Car Hire	Available	Event Risk Assessment	~~Available~~
Internal & External Contact Details	Available	Venue Derig Information	~~Available~~
Travel & Admin Handbook	~~Available~~	Packing List	~~Available~~
		IP Usage	~~Available~~

OPERATIONS

Communication

Due to the size and complexity of the project, it will be vital to have efficient communications during setup, testing and core operations.

Based on learnings predominantly via large scale FIFA and UEFA operations, FWC26 will make use of the following channels for communication:

Google Chat
- The vast majority of communication during the setup and operations periods will take place here, Groups will be set up pre-tournament
- Communication here is non time-critical or lower priority. Due to the volume of messages, users should not solely rely on “tagging” an individual if urgent or time critical.

VoIP / Phones
- Urgent, high priority or confidential communication between venue and IBC should go via the VOIP phone system

HBS / Riedel Talkback
- Talkback will typically only be used within the venue (e.g. VM to Engineering) or within the IBC (e.g. VR Op to AOT Op). Talkback over the remote link (venue to IBC) is not generally available and any comms links are “partyline” systems (ie. it is **not private** and will have many people listening in on the shared channel).

WhatsApp Groups
- Can be setup venue to venue for more social messaging. Please keep work related discussions to the Google Chat groups to ensure others can see it

Email
- Mailing lists have been setup and will be predominantly used during the build-up to the event or for sending out more detailed communications, for instance and operational or software updates.

Local eSim
All international staff onsite in the US, Canada and Mexico are eligible for an esim or equivalent. Please see Travel and Admin Handbook for more information

VOIP Numbers

Ext	Location	User
201	NYNJ	Venue Manager
202	MIA	Venue Manager
203	ATL	Venue Manager
204	DAL	Venue Manager
205	KC	Venue Manager
206	LA	Venue Manager
207	BOS	Venue Manager
208	VAN	Venue Manager
209	SEA	Venue Manager
210	CDMX	Venue Manager
211	PHL	Venue Manager
212	HOU	Venue Manager
213	TOR	Venue Manager
214	SFBA	Venue Manager
215	MTY	Venue Manager
216	GDL	Venue Manager
220	IBC Tech Ops	Duty Manager
221	IBC Tech Ops	Tech Manager
222	IBC Tech Ops	RC TG
223	IBC Track Ops	Tracking Operations Supervisor
224	IBC Data Ops	Data Operations Supervisor
225	IBC FOC	QC Duty Manager (Officiating)
226	IBC HEI Office	Project Management Team

SMART Remote

SMARTRemote will be used extensively during FWC26. It is vital that everyone is familiar with the application prior to arriving onsite.

It will be used across VAR, GLT, EPTS operations. It will also be used for extensively for replay devices.

Please familiarise yourself with:

- SRS vs SRC
- Adjusting bitrate and frame rate
- Assigning computer groups and using filters
- Difference between issues (tearing, lagging, slow response)

Note - SMARTRemote is a GPU based software. Therefore if you have an underperforming GPU or other applications are also using the GPU (specifically the nvenc encoders), the performance of SR shall be impacted

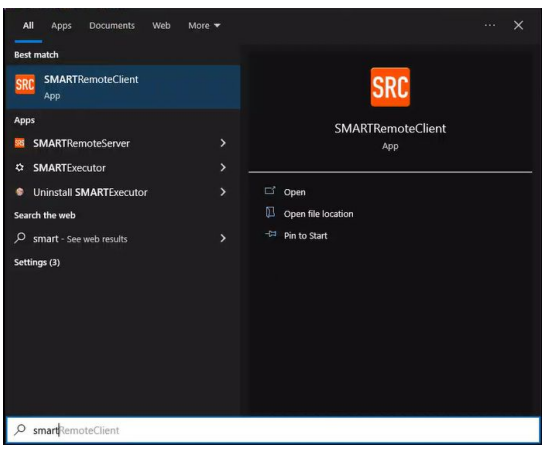
Bitrates / frame rates:
Main and Spotter Viewer monitors (refresh rates) and SRS must be set to the following

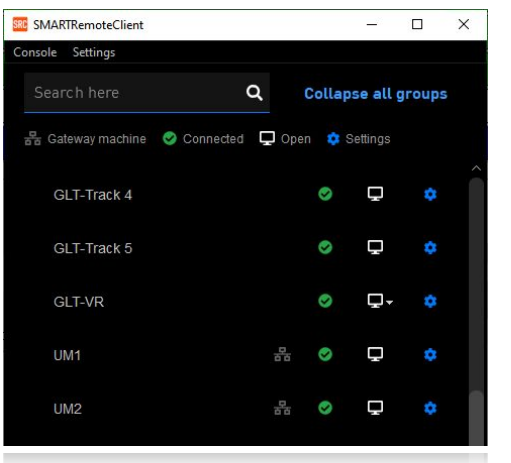
- **10Mb/s, 59.94fps**

All other SMART Remote Servers should have monitor refresh rate and SRS to be to the following

- **10Mb/s, 29.97fps**

Route connections via this server:
For DRP reasons, SMART Remote shall be implemented with SRS using the ability to route connections between the .2 and .3 networks. This ensures seamless failover between the .2 and .3 networks if the primary were to fail (or elements of it failed)





IT Security

The security of our IT systems is hugely vital at any event, let alone a tournament of this scale, complexity and importance. HEI have reviewed the IT Security based on various factors and have assessed the following measures

1. **Venue Internet**
    a. Each venue shall have 2 wired internet connection
        i. A wired internet connection shall be provided by FIFA in the FTI Equipment Room (OB Compound). Whilst this connection can be considered secure in as much as it passes through FIFA IT managed firewall, this connection shall also pass through a HEI configured firewall prior to the connection then being available to HEI staff. This connection shall be used within the system for general internet access e.g. Google chat, FTP, checklists etc.
        ii. A wired internet connection shall be provided by FIFA IT at the Media Tribune for the sole use of the Event Data Collector. This connection must only be connected to the EDC laptop. This laptop should only be used to access FELA (Event Data web application) as well as login to HEI account for Google Chat / Teams / Instant Messaging.
    b. No one other than HEI staff shall have access to these connections.

2. **IBC Internet**
    a. The IBC shall receive 6x wired internet connections, 4x provided by FIFA IT and 2x provided by HBS.
        i. Whilst the connections from FIFA IT and HBS can be considered secure in as much as it passes through FIFA IT managed firewall, these connections shall also pass through HEI configured firewalls prior to the connection then being available to HEI staff or the HEI system. FIFA 1 connection is deemed technical, and shall be used solely for the provision of live data (EPTS and Event Data). The FIFA 2 connection is considered for general internet access e.g. Google chat, FTP, uploads, checklists etc. HBS 1 connection is used for the delivery of SRT streams to DAZN Japan and is also considered a technical connection
    b. No one other than HEI staff shall have access to these connections.

3. **Software and malware security**
    a. Machines and servers shall either run ESET or Crowdstrike. ESET is starting to be phased out and Crowdstrike is taking its place.

4. **Physical Security**
    a. All areas of HEI operations are heavily restricted to accredited individuals only and cannot be accessed by members of the public. Physical security is present on all access points.

5. **Escalation and communication procedure**
    a. The venue manager shall be directly responsible for IT security at their site and shall report any IT related security issues to the PM.

Feeds

Production format is 1080p59.94 HDR (3G SDI). HDR is provided in the HLG format
Multilateral datatainment feeds - HDR (input and output).
Unilateral - SDR

Note - we handle 10-bit HDR signals transparently throughout the system. There is generally no conversion taking place on ingest or output for VAR or combined VR replays.

All outputs, at venue and at IBC must be referenced (B+B). Distribution of reference is via a Yellobrik DVA 1714 DA in the racks (onsite and IBC), and a composite video DA in the HBS datatainment rack for multilateral. Note, unilateral feeds are not expected to require referencing as handover is SRT.

PGM:
PGM is available on venue as CSF (Clean Stadium Feed). ESF (extended Stadium Feed) with permanent clock and score is also available at the IBC and should be used due to archiving requirements. ESF is on a A/E, B/F, C, D distribution.

Outputs:
Handover the following feeds at each venue:

- GLT BoxC Left 3G HDR (3G SDI)
- GLT BoxC Right 3G HDR (3G SDI)

Handover the following feeds at each the IBC (per match distribution)

- VAR Replay 3G HDR (3G SDI)
- VR Replay* 3G HDR (3G SDI)

*This VR feed is a switchable feed going to TV. The VR Operator at the IBC will remotely switch between UM Left (full frame rate replays), UM Right (full frame rate replays), GLT VR and AOT VR in a Viewer running on VR machine. This feed also includes the auto replay functionality on high confidence offside/onside events

HDR Look Up Tables (LUTs) - Converters
The FIFA / HBS approved LUT may need to be installed on SDI-HDMI converters in the venue such that HDR signals can be displayed on normal monitors with an optimised colour conversion. Use these converters on full screen monitors showing only HDR feeds. FOr the IBC, it is anticipated that all TVs showing HDR feeds shall be Hisense HDR screens and that no LUT conversion is required on the converters. The exception is the ISO feeds on GLT/EPTS monitors in the Track Ops room and any live monitors in the Data Ops room

For multiview screens showing both SDR and HDR, use converters which do not have the LUT installed

Audio Embedding:
There is no embedding of audio planned for this tournament. Riedel are responsible for any embedding of audio on feeds

Stadium Validation Match (SVM)

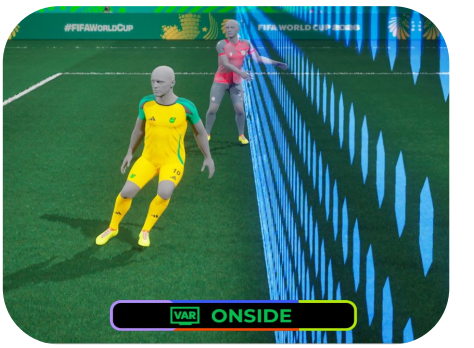
A greater emphasis is being placed on the stadium tests and all venues will have a test match

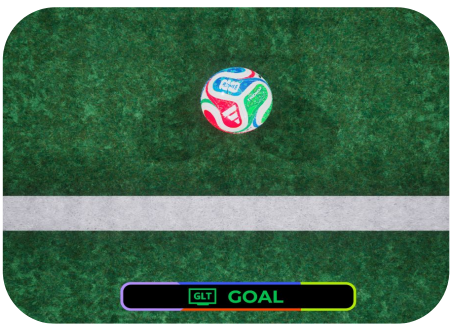
The centralised planning sheets will have the most up to date information about the FTI stadium validation match schedule in case there are further updates. This will also include the VOR / Match allocations

All technologies must be fully functional for the Stadium Validation Tests. This should be considered the first match day. These matches should be treated as if they were real games.

Elements such as the following will be tested during the Stadium Approval Tests:

- ✓ Main & Backup RRA
- ✓ Watch Messaging
- ✓ Communications (Riedel) incl al alerts (GOAL, Offside, GK encroachment) and Viewer Atmo
- ✓ GLT & Advanced Semi Auto Offside System (AOT)
- ✓ Auto VR, incl Replays to broadcast for GLT and AOT & GK line of vision
- ✓ All Replay devices, including remote injury spotter
- ✓ Local VAR station / DRP
- ✓ Live Archiving and post match clips





11 vs 11 using 2x 20 mins halves to simulate a number of incidents to check all technologies. Players will be using numbered kits for Home and Away.

QCMs will be monitoring the checklists and deliverables for this match.

FIT Tests

Labosport has been awarded the contract to certify the GLT installations ahead of the FWC26.

FITs have been scheduled with FIFA FTI who have shared the information with the stadium management team. Please check via FTI that the schedule has not changed and that the pitch management will be ready for the designated timing shown below.

Match Number	Round	Team A	Team B	Venue	Date (IBC)	Time (IBC)	Time (local)	Time (ET)
910	FIT	NULL	NULL	Mexico City	8 June 2026	20:00	19:00	21:00
913	FIT	NULL	NULL	Toronto	9 June 2026	17:00	18:00	18:00
906	FIT	NULL	NULL	Los Angeles	9 June 2026	20:00	18:00	21:00
901	FIT	NULL	NULL	New York/New Jersey	10 June 2026	17:00	18:00	18:00
907	FIT	NULL	NULL	Boston	10 June 2026	17:00	18:00	18:00
908	FIT	NULL	NULL	Vancouver	10 June 2026	20:00	18:00	21:00
914	FIT	NULL	NULL	San Francisco Bay Area	10 June 2026	20:00	18:00	21:00
911	FIT	NULL	NULL	Philadelphia	11 June 2026	17:00	18:00	18:00
904	FIT	NULL	NULL	Dallas	11 June 2026	18:00	18:00	19:00
912	FIT	NULL	NULL	Houston	11 June 2026	18:00	18:00	19:00
902	FIT	NULL	NULL	Miami	12 June 2026	17:00	18:00	18:00
903	FIT	NULL	NULL	Atlanta	12 June 2026	17:00	18:00	18:00
909	FIT	NULL	NULL	Seattle	12 June 2026	20:00	18:00	21:00
905	FIT	NULL	NULL	Kansas City	13 June 2026	18:00	18:00	19:00

MTY and GDL FITs were completed in March during the POTs.

Details of the lead test engineer for Labosport will be communicated in due course.

All accreditation for Labosport has been submitted and accepted.

As normal, 6x watches should be handed over to Labosport for the FIT.

Dallas stadium will be using the FWC Finals ball for the test. Given the ball is not yet publicly known, you must keep it under wraps until it's needed.

As with match operations, FITs will be staffed with an operator at the IBC. The person leading each FIT will be included in the detailed shift planning which will be available closer to the time.

Sync Test

Overview:

- TTG should conduct this process, under supervision of VTG and understanding of the VM. The VM is ultimately responsible for the Sync test taking place at the correct time
- The sync test will be scheduled in the *Countdown to Kick-Off* or equivalent official document issued by the FIFA Venue Team.
- Ensure that TTG is pitch-side 5 minutes before the start of the sync test time
- **Pans & Calibration must be done before Sync**
- Ensure that the VAR system is recording all cameras. Coordinate this with the IBC VOR
- Contact the QCM (#201) to inform them you are ready for sync at -5mins

Procedure:

- Start at end 1.
- At the start of the sync test time step onto the pitch, in front of the goal and in view of all cameras. Hold the ball up in the air to get the attention of all camera operators.
- Await the go ahead from VTG which will be given once all cameras are focussed
- Bounce the ball on the floor whilst turning around in a circle to ensure every camera can see the ball bounce. Continue this for around 15 seconds or until the VTG confirms its complete
- Step fully off the pitch and walk around to end 2 on the broadcast side. Bounce the ball every 5m or so
- Repeat the above steps for end 2 when given the go ahead by the VTG
- **A fast paced shot into the goal at each end after ball bounce.**

  This can only be done once the 4 broadcast cameras have FOV (need to have green calibration on viewer as well)

Never walk across the pitch from end 1 to end 2! Remember all HB cameras are focussed on you and these images are being distributed to key stakeholders.

Always act professionally

Communications:

- The VTG and TO2/ARO2 should communicate via mobile phone. The TO2/ARO2 (pitchside) should use headphones
- The RCTG, VTG and HB can communicate, if required, via the technical partyline. Note - this is not a private communication channel

Once complete, the RO should synchronise the VAR system as normal

VENUE

Onsite Cabins

Similar to previous FIFA tournaments, all technical equipment shall be based in a dedicated FTI Equipment Room. This will be a double size cabin (i.e. 6m x 4.8m). This allows us to separate operations away from equipment.

All onsite monitoring will take place in another double cabin (FTI Ops Room). All staff will be located in this area and will monitor from here. The FTI Operations Room is solely for HEI use.

All cabins will have A/C units, power feed, power distro box and cable entry flaps. Wall sockets in the cabin are for domestic equipment only - **do not power technical equipment from these wall sockets!**

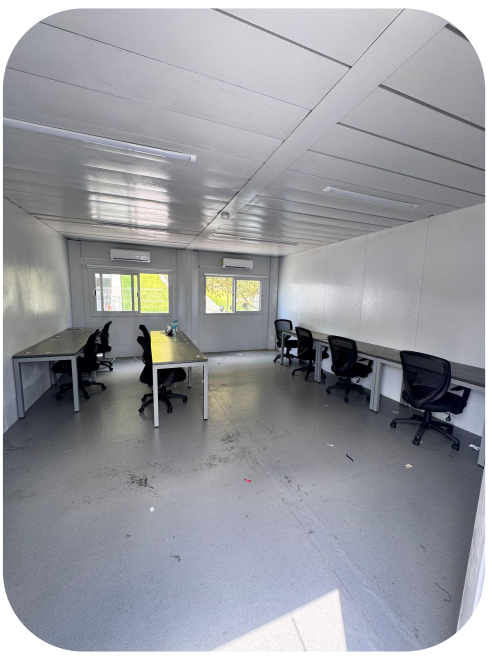
HBS provide power as NEMA-20R connectors, with 120V (AC). These are labelled RED and BLUE. Both supplies are technical power, and equipment should be distributed across both sources

Due to reduced voltage, **all servers must plug directly into the NEMA power socket, and not via a PDU**

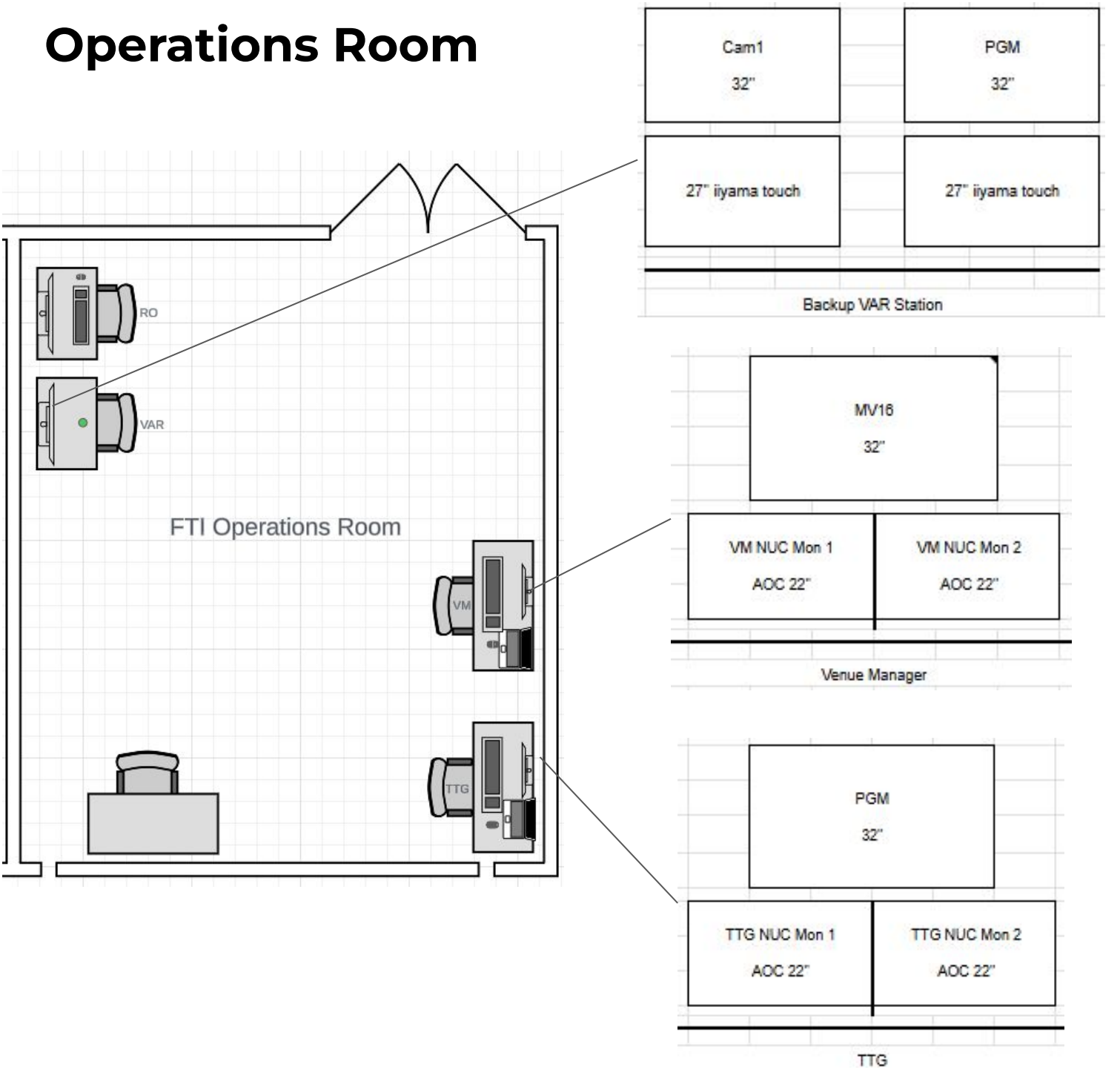
A break area shall be located by in the compound. Please bring reusable water bottles / coffee mugs. Every effort should be made to **not** use single use plastic bottles/coffee cups.

Cabins should all have a suitable number of desks and chairs. Use the single desks for the backup RO/VAR station

The VM should speak with the FTI Venue Manager onsite if there are any questions.



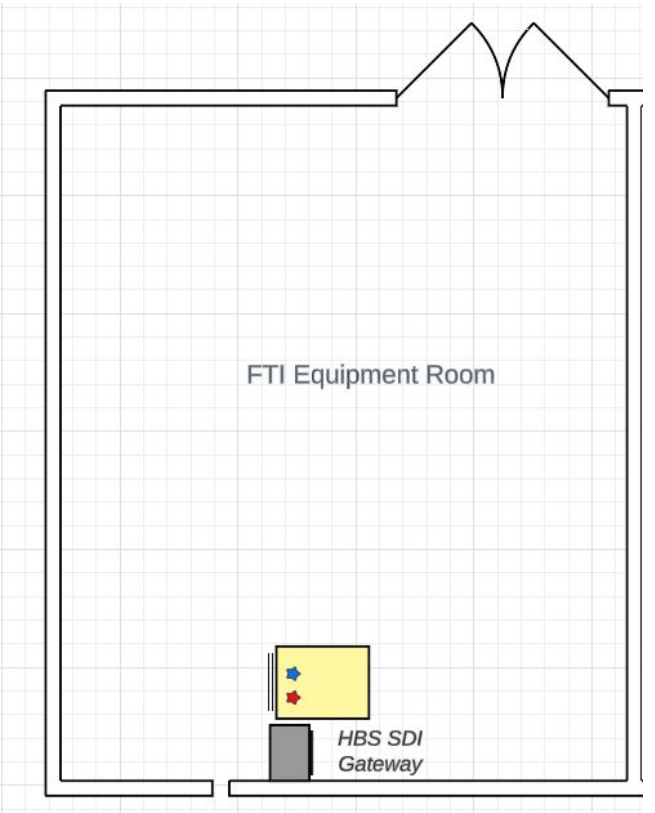
Operations Room



- Only small form factor machines, laptops and converters etc are installed in the ops room
- Backup VAR Station - RO and VAR only. Use 2x single desks for this backup station
- Double desks for VM, TTG and spare

  Depending on the exact placement of the door, the orientation may need to be adjusted.

FTI Equipment Room (Server Room)



Example from FIFA Women's World Cup 2023 - June 2023. *Not fully representational for FWC26*

- 1x 28U rack for HEI. Other equipment may include the SDI gateway (from HBS), power distro, fibre termination / enclosure, kinexon rack, Riedel equipment
- Depending on the exact placement of the door, the orientation may need to be adjusted
- Ensure sufficient room at rear and front of rack
- Room is to be shared with other FTI Providers (e.g. Kinexon and Riedel)
- FTI Equipment Room is not for general storage! Keep the room neat and tidy, free from food / drinks

Equipment Racks

There will be 1x rack per venue. These have all been custom built for the FIFA World Cup 2026. The rack will contain 3x servers, which will run the backup VAR feed capture, viewer, as well as provide redundancy for the applications which relay the tracking camera streams to the IBC.

Rack Photos - Pre Shipping

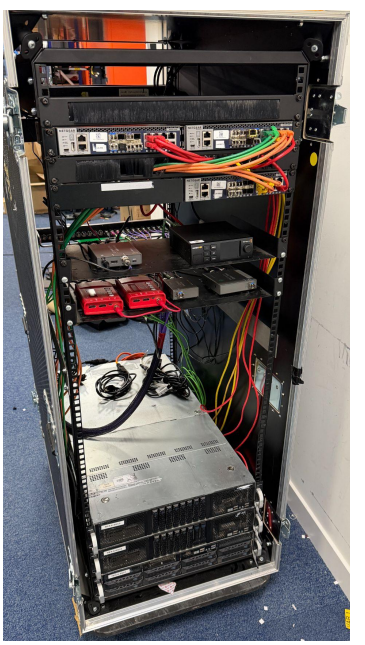
To do onsite:

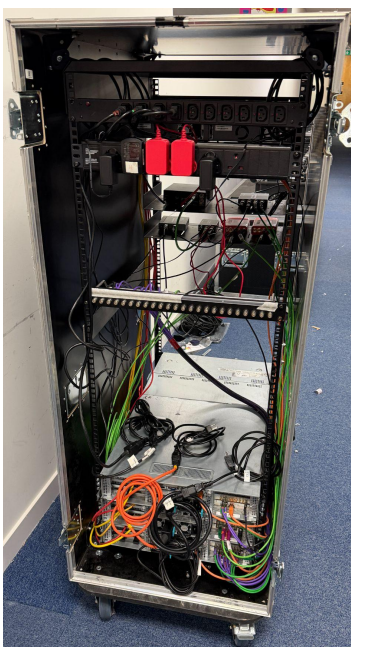
- Connect up power
- Patch in network for HEI Ops room
- Patch in SDI (6x backup camera feeds and 3x feeds to HBS)
- Patch in fibre for cabinet network switches and watches (GLT and AOT Z-Cams & GLT Watches)
- Patch in fibre for RRAs and CDP for replay devices / media tribune
- Patch in Kinexon PTP

Notes:

- Keep hot area clear at rear
- Use correct strain relief cable bungees / webbing (not cable ties!)
- Ensue everything is correctly labelled.
- Leave adequate space behind racks for access and airflow
- Avoid cable spaghetti!

Ensure the FTI Equipment Room and our racks look professional and presentable at all times





Network

Venue-IBC:
HBS will make available 2x 5Gbps links between each venue and the IBC. The links must negotiate to 10Gb. These will be handed over as duplex LC connection (i.e. 2 pairs). Ensure this connection along with relevant patch cables are protected and secure.

We are running a .2 (primary) and .3 (secondary) setup..

The connections will be live 24/7 and shall be configured using VLANs. There are 6x VORs and as such venues will not always be connected to a VOR, however they will always have connection to the Gateway NUC at the IBC.

There are 6x GLT stations and 6x EPTS Tracking stations at the IBC. These will generally be matched to a VOR, though during setup/FITs they may be separated

IBC shall be running network management software which monitors the links from each venue. It also checks things like link status, port flapping, bandwidth etc. for more enhanced support and monitoring. As a result, unless authorised by the IBC, ***you should not be changing the configuration of any switch***

Venue Switches - Ports:

- The network switches are not configured as a stack, but will be connected to one another as per normal setups.
- All core switches are Netgear M4300s
- It is vital that the switch ports are correct and connected correctly. This ensures that every venue is standardised, and assists with identifying any flapping or issues on a port.
- Switches have had a base configurations loaded.
- The trunk port that connects back to the IBC is critical. Never change the port unless explicitly approved by the IBC team.
- All other ports have had relevant PVIDs and VLANs set and so do not change what device the port is meant to be connected to.
- Due to the way the network has been configured, you will not be able to access or ping the switch using any of the ports. All configuration must be completed from the IBC.
- If you want to check connection to the IBC, you can do so by pinging your gateway NUC. This will be online 24/7 and irrespective of whether your venue is connected to a VOR, rack or GLT/EPTS station.

Venue Switch Plug Up

Network continued

Internet
Internet has been requested as a single wired connection with up to 100Mb up/down in the FTI Equipment Room, and another single wired connection at the Media Tribune for Event Data Collection. This should have all necessary ports open, such as FTP, Splashtop, FELA, EPTS Cloud Uploader etc

Internet in the FTI Equipment Room must be connected via our Fortinet Firewall. The internet at the Media Tribune (for EDC) can be connected directly into the laptop

As with all internet access at FWC26, strict Sony IT security procedures will be in place. Only permitted machines must be on the internet.

PTP Connection
PTP connection shall be provided by Kinexon and handed over to HEI in the FTI Equipment Room onsite. This is in line with the work flow at previous FIFA events.

HBS may also provide a PTP connection however this shall not be required.

The onsite TTG, with supervision from the IBC, shall be responsible for connecting this into the relevant network switch.

SDI Feeds

You only need to pickup 6x feeds from HBS, used as part of the DRP backup VAR/VOR.

You will also be providing GLT L and GLT R cams to HBS from the onsite system, along with the VAR output from the backup viewer. This backup feed may not necessarily be plugged up to TV

Video feeds to be connected to the SDI gateway provided by HBS in each FTI Equipment Room.

You will have a short loom to enable you to do this (~3-5m)

Host Broadcaster - HBS

The Host Broadcaster is HBS. You may be familiar with working with them on previous FIFA tournaments.

For FWC26, the production will be hybrid, with certain functions onsite and others at the IBC.

The onsite technical facilities providers will vary from venue to venue.

HBS Code	Venue Number		Technical Set Up	Facilities Provider
NYNJ	1		NEP	OB TRUCK
MIA	2		NEP	FLYPACK
ATL	3		NEP	OB TRUCK
DAL	4		EMG	FLYPACK
KC	5		NEP	OB TRUCK
LA	6		EMG	FLYPACK
BOS	7		AMP VISUAL TV	OB TRUCK
VAN	8		DOME	OB TRUCK
SEA	9		EMG	FLYPACK
CDMX	10		EMG	FLYPACK
PHL	11		GAME CREEK VIDEO	OB TRUCK
HOU	12		NEP	OB TRUCK
TOR	13		DOME	OB TRUCK
SFBA	14		NEP	FLYPACK
MTY	15		MEDIAPRO	FLYPACK
GDL	16		MEDIAPRO	FLYPACK

Feeds - Venue to/from IBC

Venue to IBC (streamer)

MV4 quad split containing
- Main RRA loop return
- Backup RRA Loop return
- Ops Cam
- [TBC]

IBC to Venue (hardware encoder/decoder)

- Backup circuit for RRA (via hardware encoder/decoder)
- MV16 (via streamer)

Multiview 16

A 16-way multiview feed will be streamed from the IBC to venue. This can be used to monitor all key outputs and selected inputs. This should be shown on the 32" monitor on the Venue Manager desk

The 16-way multiview is generated at the IBC.

RRA Feed (to Venue)	GLL	VR Out	INFO
BRRA Feed (to Venue)	GLR	VR (No Priv)	MIS
MV OUT	VOR CAM	PGM	STN CAM
SV OUT	VENUE MV4	CAM1	RO MON

Talkback

We have requested 1x talkback unit for HEI in the FTI Operations Room
- VM: For coordination of pans, syncs, camera feeds etc as normal

Riedel will also be installing talkback units in the Ops Room in the RO/VAR backup positions. These are not to be used under normal circumstances

Its likely that some limited communications will be provided between Venue and the IBC, however like previous World Cups, this may include the use of a “Partyline” - this is a public channel that is open to multiple parties. It is not private and anything transmitted is heard by many people

Power

HBS will supply 2x BTP (Broadcast Technical Power) supplies, A and B, to each room. This is instead of the typical 1x BTP and 1x BDP (broadcast domestic power) supply. Both BTP supplies should run continuously throughout the tournament. Connections are 120V via NEMA-20R connectors

FWC26 Power (HEI Use Only)

Area			Type	Voltage	Connector	Quantity	Note
Per Venue	TV Compound	FTI Equipment Cabin	BTP - A	120V	NEMA-20R	5	
			BTP - B			5	
		HEI Operations Cabin	BTP - A	120V	NEMA-20R	2	
			BTP - B			2	
	Venue Side	GLT/SAOT Cabinets	Venue Specific Stadium Power via UPS or BTP	120V	NEMA-20R	1x 10Amp at each cabinet position with a power strip of 3 connections	The number of positions as indicated on installation plans
		Pitchside Positions	BTP	120V	NEMA-20R	1	RRA
			BTP	120V	NEMA-20R	1	BRRA
			BTP	120V	NEMA-20R	1	Pitchside Tablet Position 1 - Team A Bench
			BTP	120V	NEMA-20R	1	Pitchside Tablet Position 2 - Team A Bench
			BTP	120V	NEMA-20R	1	Pitchside Tablet Position 1 - Team B Bench
			BTP	120V	NEMA-20R	1	Pitchside Tablet Position 2 - Team B Bench
			BTP	120V	NEMA-20R	1	Pitchside Tablet Position - Medical Team Seating
		Media Tribune	BTP	120V	NEMA-20R	1	Media Tribune Fibre Drop Position(s) - see extra notes
			BTP	120V	NEMA-20R	1	Replay Device (per position)
			BTP	120V	NEMA-20R	1	Replay Device (per position)
			BTP	120V	NEMA-20R	1	Event Data Collector

Extra Notes:
(1) In LA, 1x 120V NEMA-20R is required in each of the TEAM Booths, 1 for Team A & 1 for Team B
(2) In ATL & TOR, 1x HEI SAOT/GLT Cabinet is located at the Media Tribune FTI Fibre Drop location. At this position, it's important power requirements are treated separately (i.e 1x MT & 1x GLT/SAOT with 3way power strip)

Power

Power Procedure - Overnight / Non Match Days:
There is no need to power all the equipment / servers down overnight. The technical power shall remain online 24/7.

Given the short nature of the tournament, once the servers are booted and functioning, they should not need to be powered down.

Other technical equipment such as network switches, converters, NUCs* etc should also remain online throughout.

**It may be beneficial to restart the NUC every 3-5 days, however they should remain powered on when the team are not onsite.*

<u>All monitors should be turned off to avoid image / display "burn in"</u>

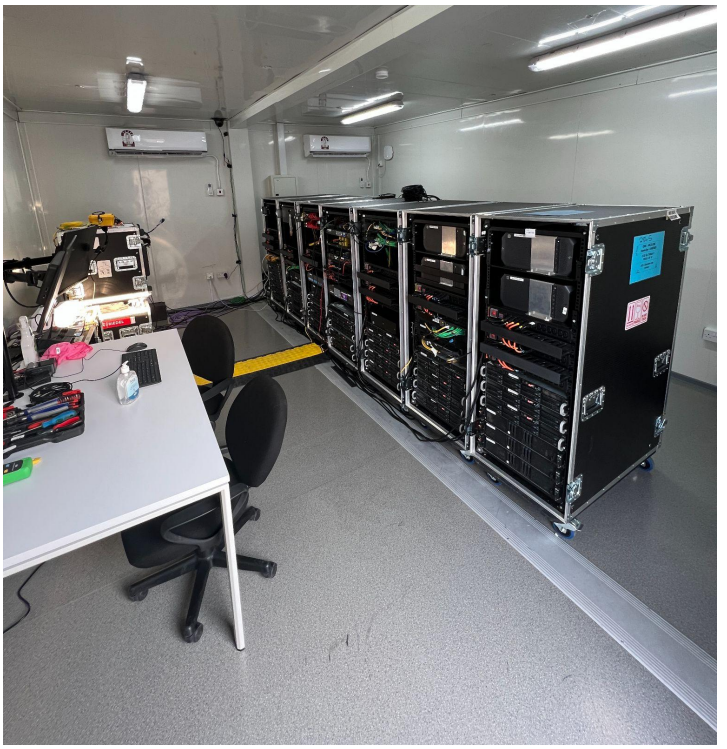
Rigging

It should go without saying that it is essential that the system is rigged neatly and professionally. This creates a good impression and perception of Hawk-Eye and the venue team. You should follow the setup guides so that all venue setups are identical
General good practice must be followed:

- **Labeling** cables (e.g Power, cables between rooms, general incoming and outgoing cables)
- **Cable Slack** - try to leave slack outside of the rooms rather than coiling inside and making it look untidy. E.g cable looms to TV, fibre to GLT cabs etc
- **Strain relief** - all cables must have adequately strain relief. Cable bars have been installed in the racks and bungees / webbing have been packed in the kits. This is especially important for fibre (due to fragile nature of it) and BNC looms (due to weight pulling on the BNC connectors. Strain relief will also allow a better view of the patch panels
- **Cable management** - This includes use of velcro ties, tape and the like to ensure a clean and professional approach to cabling. It also avoids feet and the like getting caught in a cable loop and pulling on something critical. We do not want to see 'cable spaghetti'
- **Protection** - ensure that key connections are well protected - e.g. fibre patch cable that carries the IBC traffic, RRA fibre etc
- **Floor space** - keep the floor space at the rear of the racks as free and as clear as possible. If needing to run cables on the floor, make sure they are run together and secured to the floor. Use cable matting or cabe ramps to add further protection.
- **Tidy work area** - this includes disposing of rubbish, keeping personal bags to one side, and not having kit and equipment strewn across the room/floor.

Poor rigging will cause extra issues in your setup. The VM is ultimately responsible for the professionalism of the setup

Examples from FWC22
Clean & Tidy Working Environment!



Cabin layout not fully representational for FWC26

IBC

IBC

The IBC (International Broadcast Centre) is where HBS, the host broadcaster for FWC26, as well as all the unilateral rights holders for different countries and various other media and journalists, are based.

Since 2018, the IBC is also where we deliver officiating technology from for tier 1 FIFA tournaments.

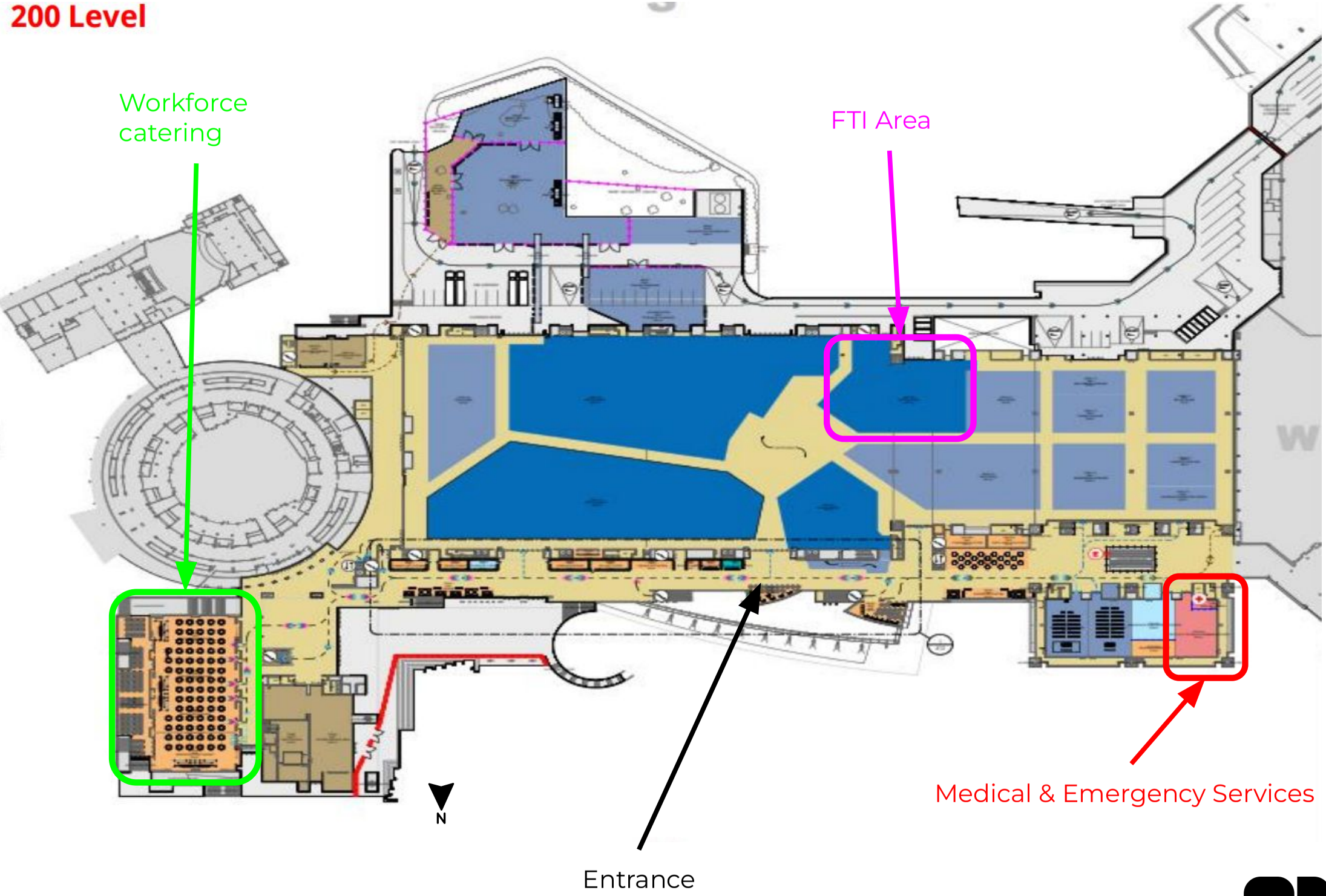
The IBC for FWC26 is located in downtown Dallas, at the Kay Bailey Hutchison Convention Center. HBS are responsible for designing, building and operating the IBC within the convention centre, and builds commenced mid January. A Pre-Rig of HEI equipment was completed in May to further prepare our operations ahead of June

For this edition of the World Cup, the following key deliverables will be located at the IBC:

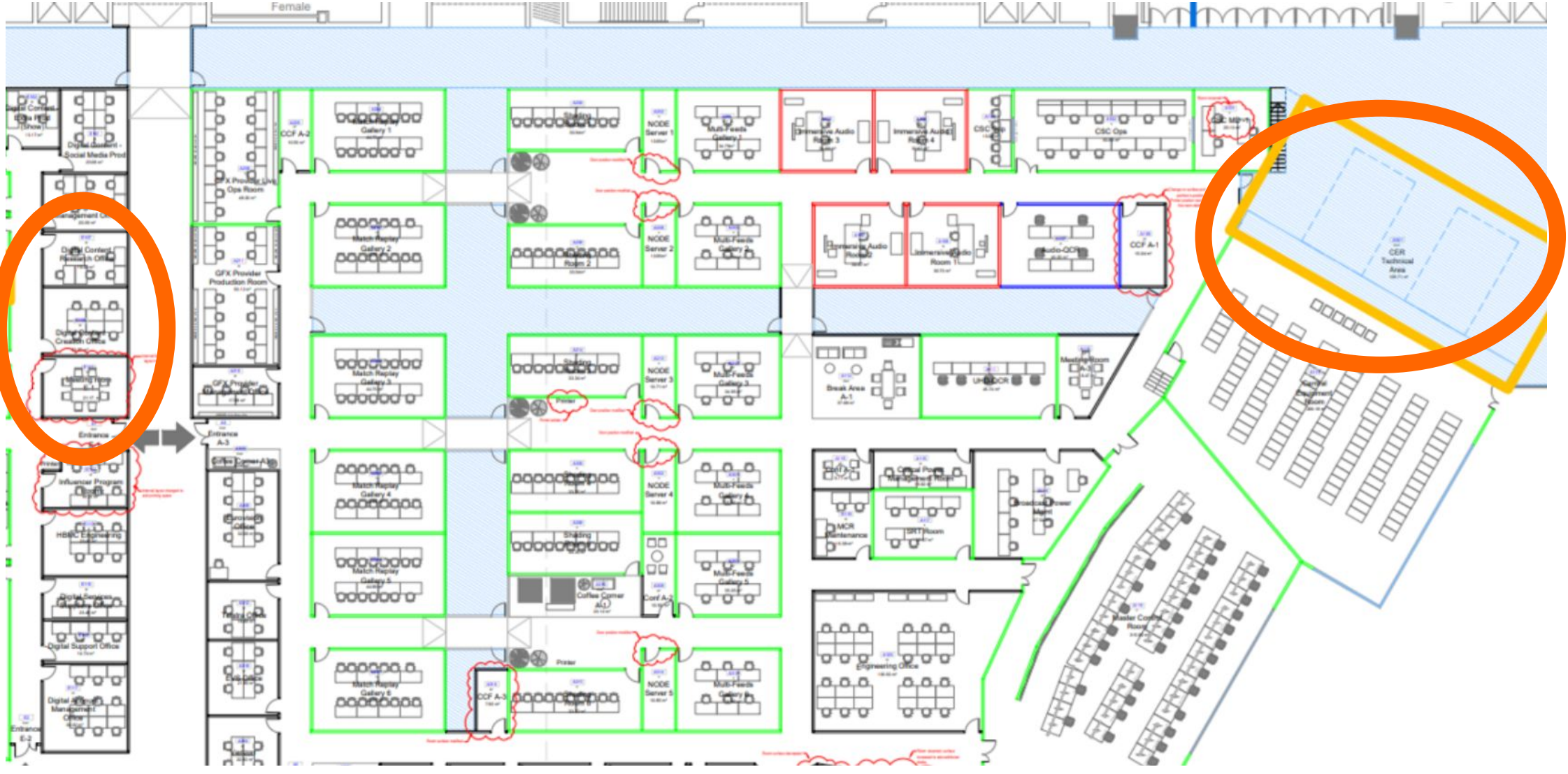
- ***6x match racks, each rack:***
    - ***VAR Feed Capture & Viewers***
    - ***GLT tracking servers***
    - ***EPTS/AOT tracking servers***
    - ***Viewer devices for medical / analysts at venue***
- ***Centralised GLT Operations***
- ***Centralised EPTS/AOT Operations***
- ***Centralised VAR Operations***
- ***Centralised medical at IBC***
- ***Datatainment (LiveAR) services - multilateral and unilateral feeds***

HEI are responsible for installation all equipment, servers, converters and monitors within our areas in the IBC. We are also responsible for cabling between our rooms, of which the majority was completed during a Pre-Rig period

200 Level


Workforce catering
FTI Area
N
Entrance
Medical & Emergency Services

HBS Datatainment Areas



FTI Area

FTI EQUIPMENT ROOM
RIEDEL
HEI Project Office
Referee Meeting Room
VOR 1
VOR 2
VOR 3
VOR Storage
Riedel
Events Mgmt
Network
TM
DM
RO QC
VOR 6
VOR 5
VOR 4
RC TG
VOR 3
VOR 2
VOR 1
VOR 4
VOR 5
VOR 6
FTI Office
FOC
Centralised Medical Spotters
Referee Analysis
FTI Meeting Room
DATA OPS
TRACK OPS
Hot Desks
GLT5
EPTS5
GLT6
EPTS6
EPTS4
GLT4
EPTS3
GLT3
EPTS1
GLT1
GLT2
EPTS2

HEI Rooms - FTI Area

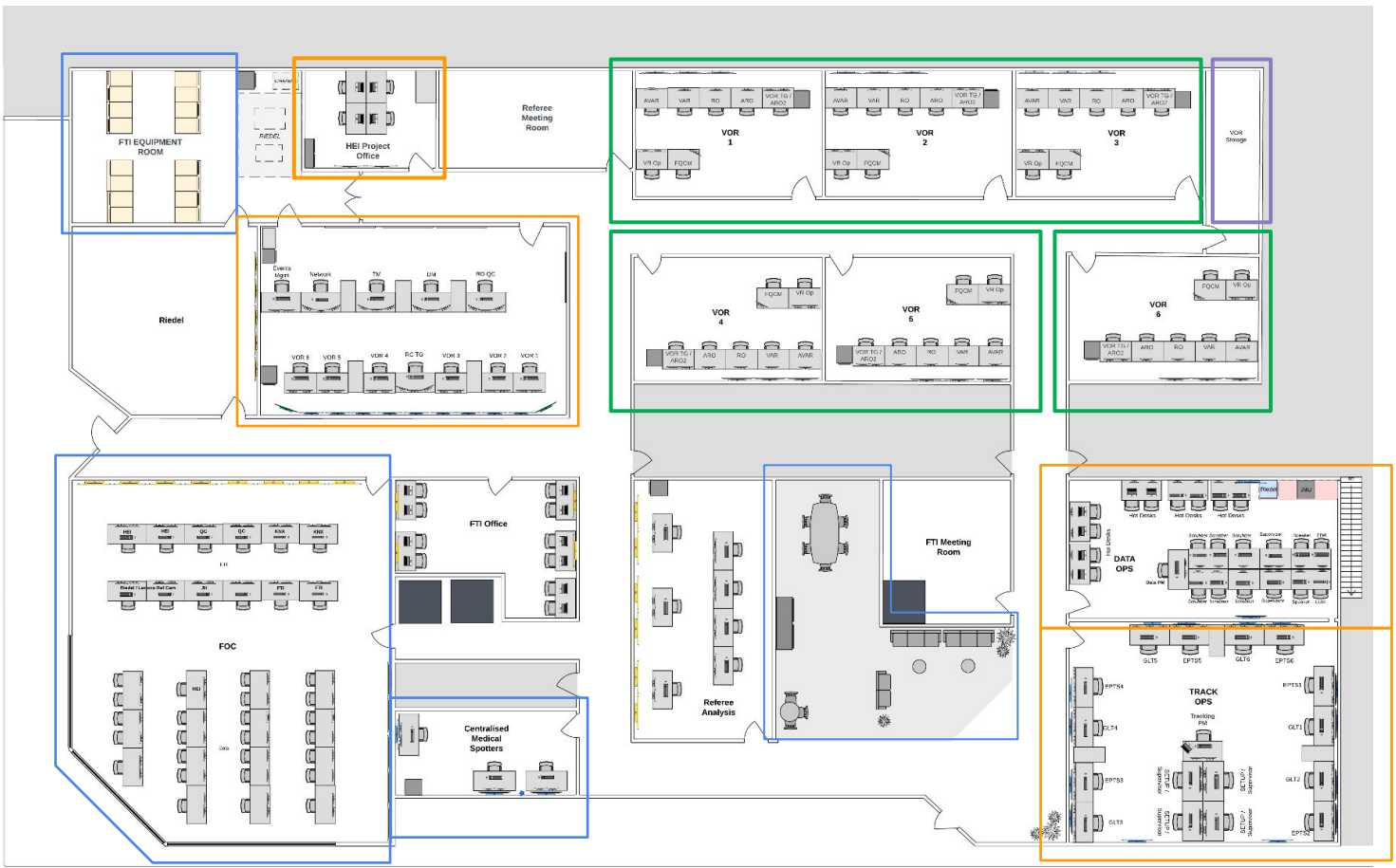
Given the size of the delivery at IBC, HEI have access to the following spaces / rooms:

- **FTI Equipment Room** - shared space, predominantly with Riedel
- **VORs (x6)** - shared space
- **Technical Operations Room**
- **HEI Office**
- **Tracking Operations**
- **Data Operations**
- **FOC** - shared with FIFA and other FTI suppliers
- **Storage Room** - shared with Riedel
- **Centralised Medical** - equipment only
- **Break Area** - shared space*
- **Datatainment (HBS area)**

Orange indicates rooms dedicated to HEI

All spaces are kept neat at all times!
It is everyone's individual responsibility to ensure they clear up after themselves.

Please note, the break area is for all FTI and Refereeing, not just Hawk-Eye. Be aware of others using the area and keep it clean and tidy



IBC FTI Equipment Room

Technical Scope

All our technical equipment shall be located within the FTI Equipment Room. This includes all VAR capture, GLT & EPTS/AOT processing, network, video routers and multiviewers, scrubbing machines, remote viewers and gateways

The FTI Equipment Room will be the main handover between HBS and HEI, and HEI and Riedel.

Riedel will also be located within this room

Rack Configuration

All our equipment is split across 6x match racks (A-F), a core rack and an additional rack.

Infrastructure

Purpose built cable support trussing within the room is installed to improve cable management.

Access will be limited to authorised HEI staff only

IBC Racks during builds and testing in Basingstoke Office

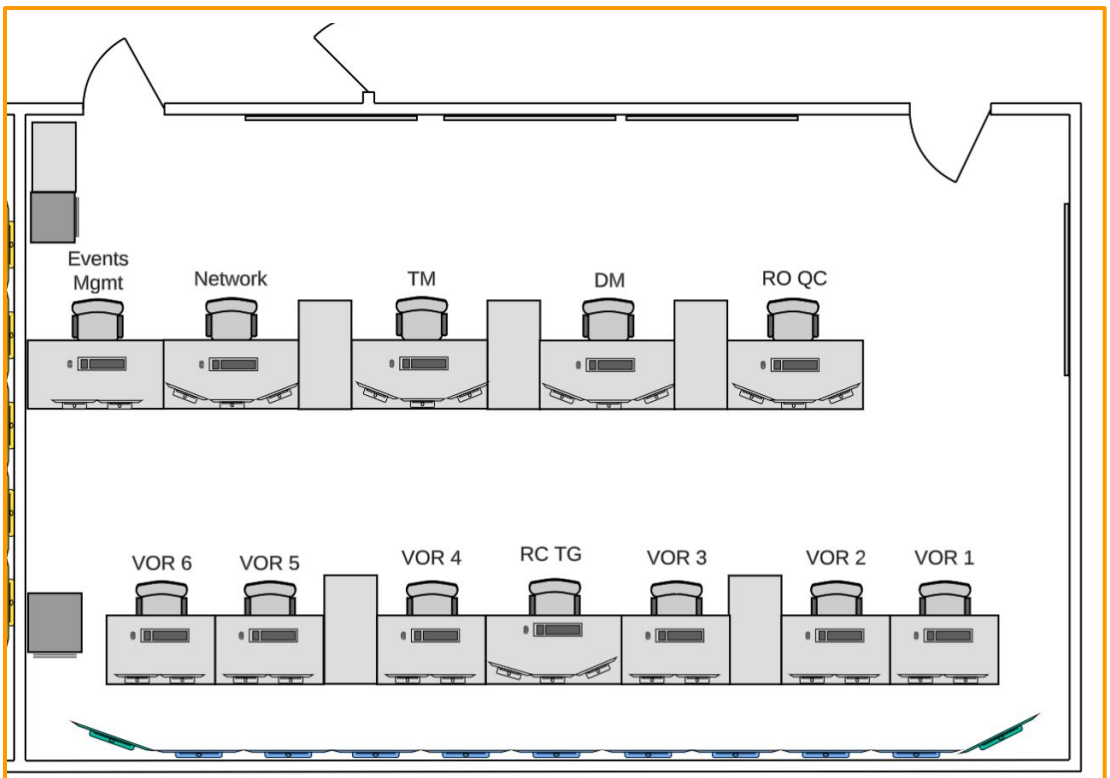
IBC Technical Operations Room

All technical monitoring of signals coming in, out and being handed over will take place in the HEI Technical Operations Room. This includes any communication with HBS MCR

The IBC VOR TGs will also be located in this room, align with the IBC RCTG Supervisor.

The on shift Duty Manager, Technical Manager and Network Manager will oversee operations from this room. During live matches, the RO QC will also monitor the quality of replay operations and log feedback as required.

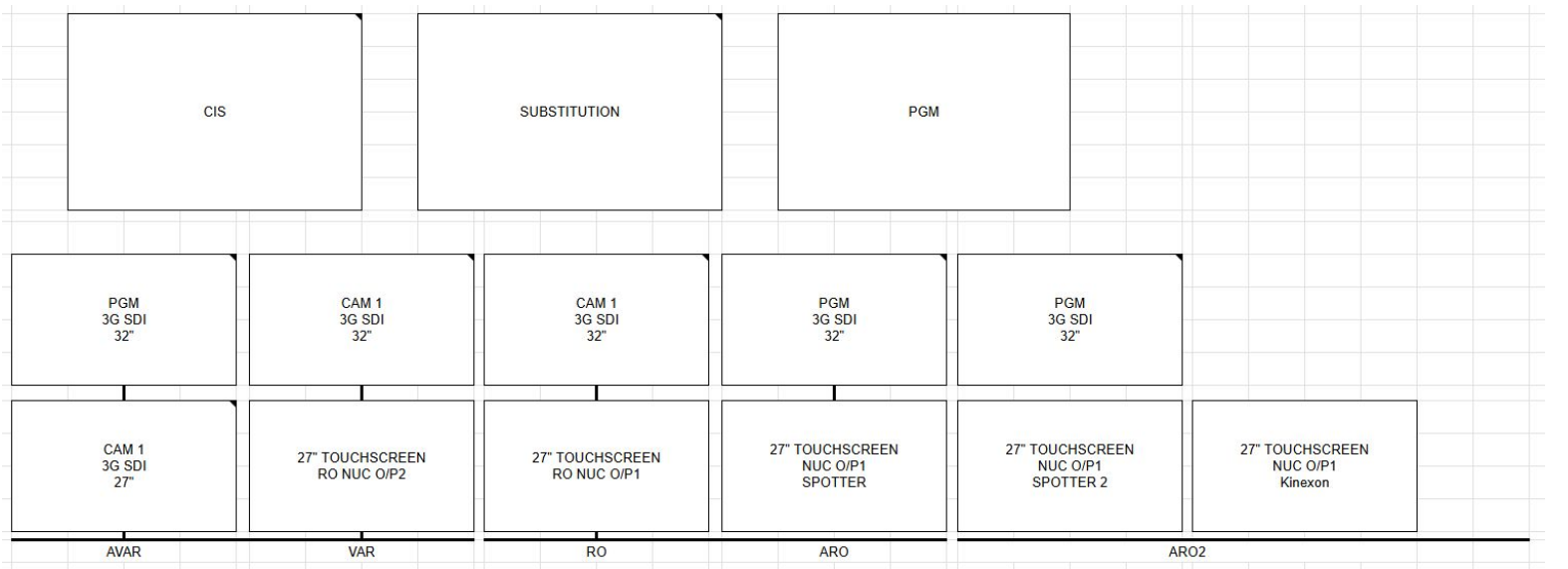
The TVs on the monitor wall shall be provided by Hisense. HEI are responsible for feeds and mounting of the TVs

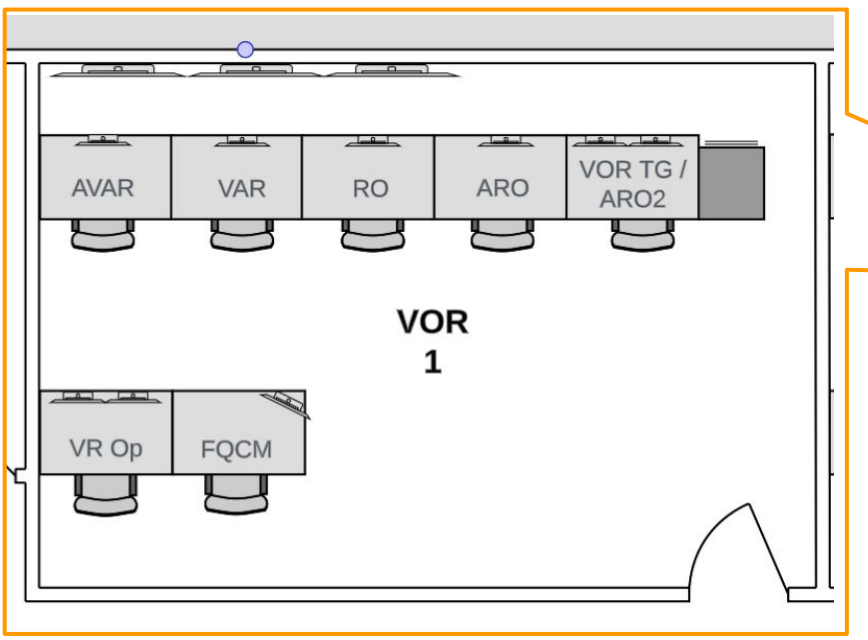


Given the technical nature of this room, access will be limited to authorised HEI staff only

IBC VORs (x6)

- No offside station
- Extra TV screen for subs. Large wall mounted TVs for PGM and CIS provided by Hisense. Feeds provided by HEI
- ARO2 is will be the eyes and ears in the VOR and will be on talkback with the VOR TG in the TechOps room
- Dual touch screen for ARO2 Spotter (to control Kinexon viewer)
- All cabling must be extremely neat and all connections labelled
- All VORs to be rigged exactly the same
- Tablets outside will indicate when a VOR is in **LIVE MATCH OPERATIONS**





IBC Tracking Ops Room

A Tracking Operations room has been assigned to HEI, and will house GLT operators, EPTS Tracking operators, Tracking Supervisors. This room has limited space and therefore cannot accommodate those who are not actively working on tracking operations

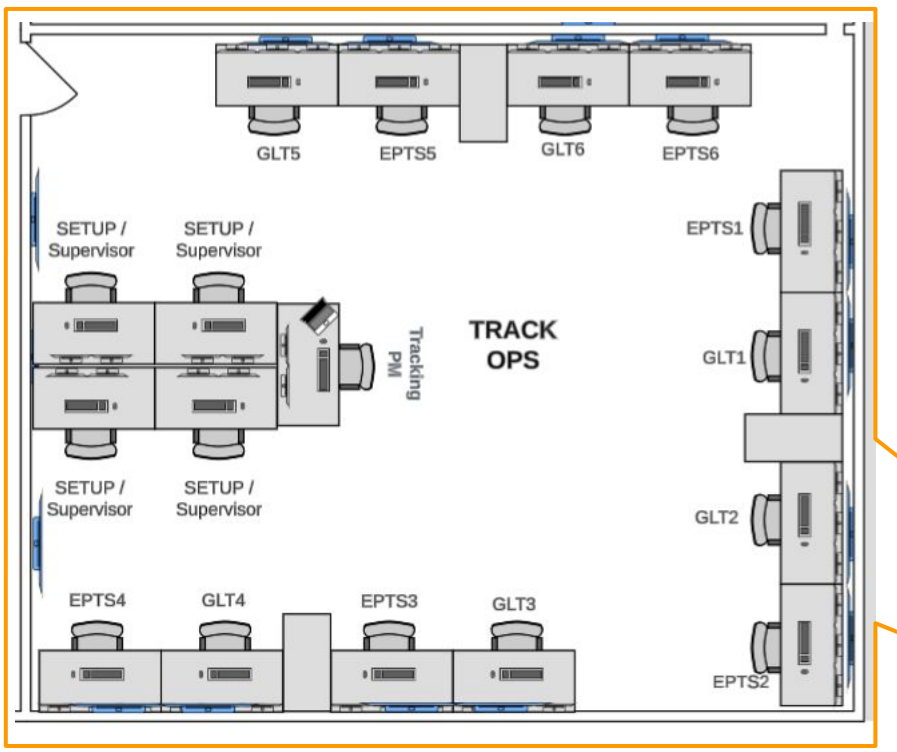
We will be continuing to operate GLT, SAOT and EPTS centrally.

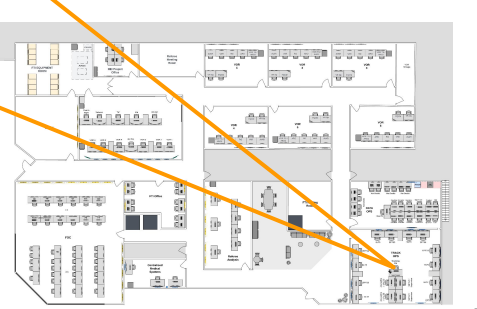
- 6x GLT stations + 2x setup stations
- 6x EPTS Tracking stations + 2x setup stations
- 1x Tracking PM Station

Centralised operations for GLT and EPTS will encompass all tasks that have previously been completed onsite such as calibration, software configuration, match operation and data archiving.

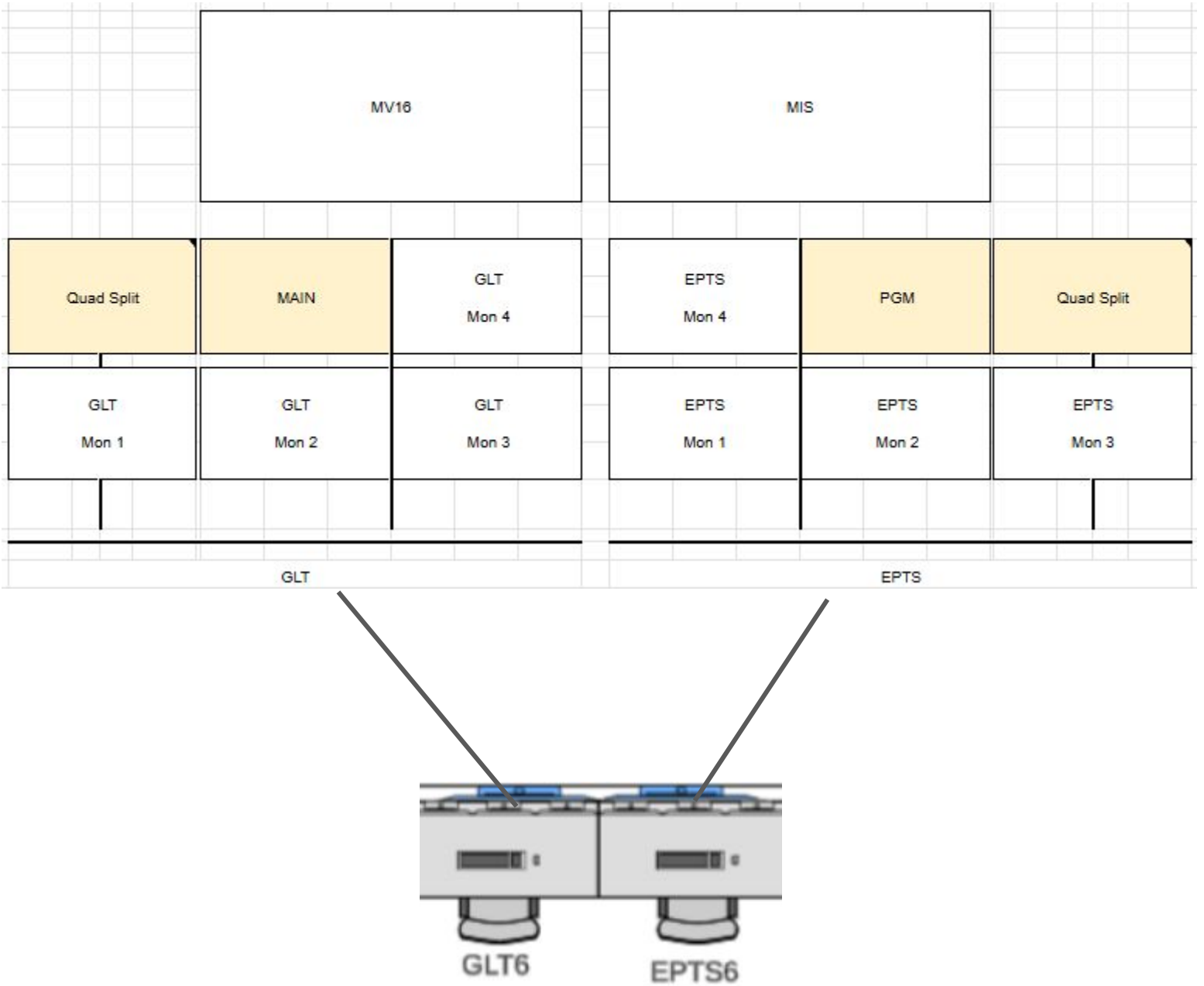
Riedel will be providing talkback panels (1x GLT and 1x EPTS per station) to allow communication within the IBC. This communication channel does not extend to the venue and therefore alternative means of communication ot venue teams should be used (Google Chat, VOIP etc).

- Large TVs per station provided by Hisense and mounted by Hbs. Feeds for all TVs provided by HEI





IBC Tracking Ops Room

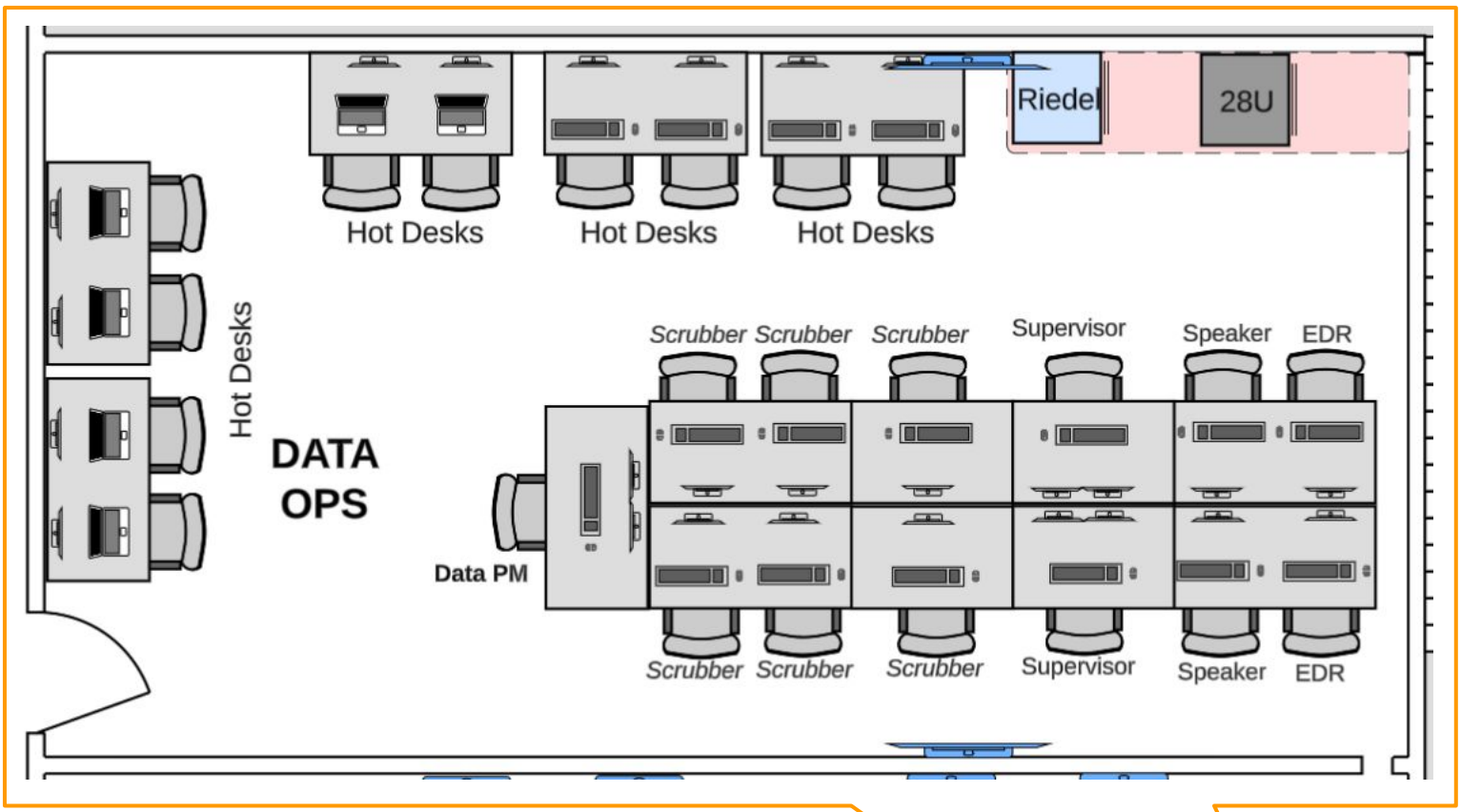


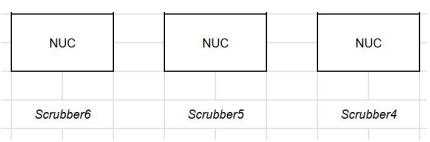
Notes:

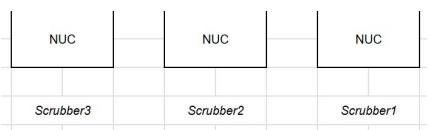
- Yellow monitors will be live SDI feeds.
- TVs on wall provided by Hisense and mounted by HBS
- 1x NUCs for GLT & SAO, each capable of running 4x monitor
- 2x Riedel talkback panel (1x GLT, 1x EPTS)

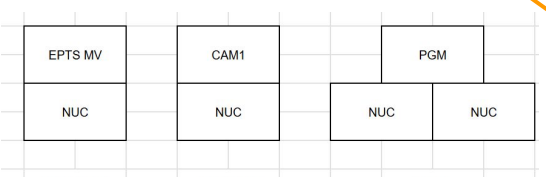
IBC Data Ops Room

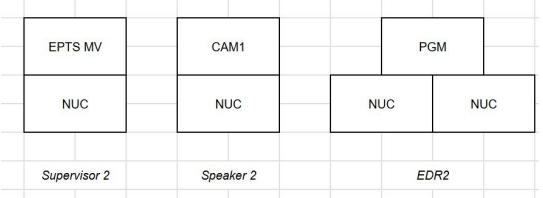
- All Data operations (excluding onsite EDC) will be conducted from this room
- Hot desks available in this area for support staff, 4x NUCs + 4 Monitors for laptops
- Large TVs per station provided by Hisense and mounted by HBS. Feeds for all TVs provided by HEI









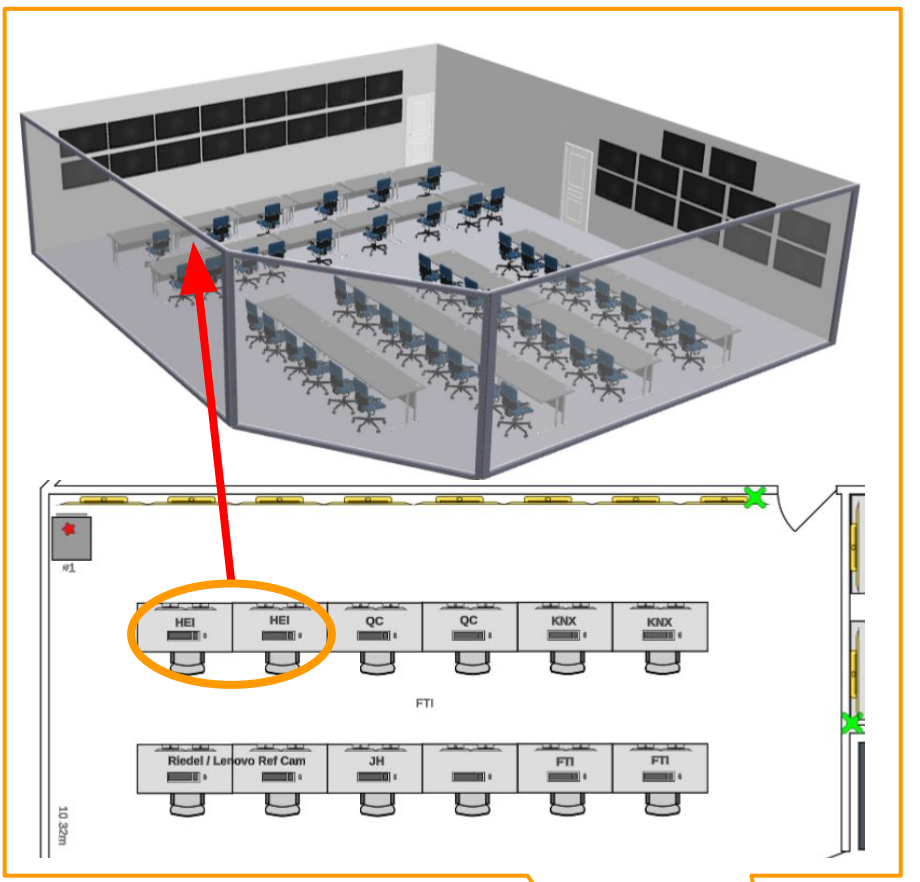


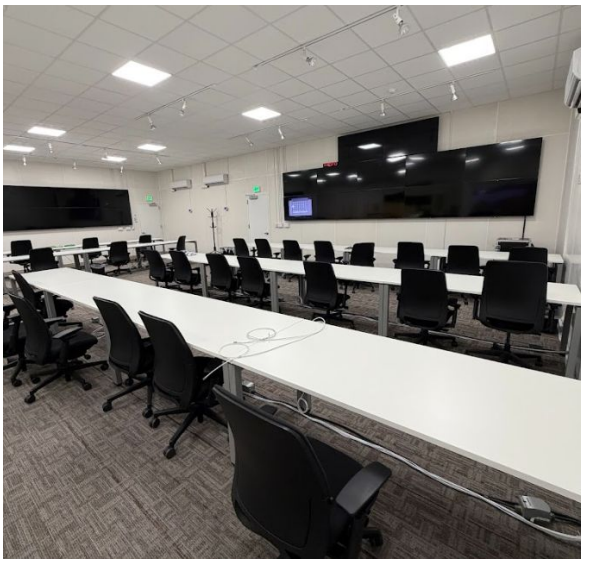
IBC FOC - Football Operations Centre

The FOC is an FTI room where they will be monitoring all data and officiating

Similar to previous competitions HEI will be providing feeds to this room (26x SDI), as well as having desks in the FTI side (for FTI QCM) and the data side (for Data QCM)

Note - HEI are responsible for providing 4x SDI feeds to Kinexon. These can be switchable by a routing panel provided to KNX.





IBC - Referee Analysis Room

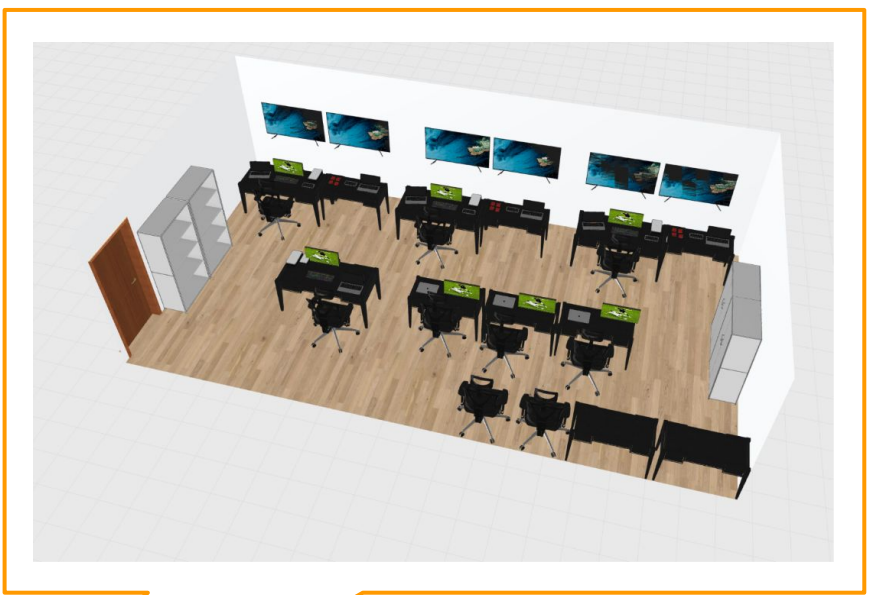
Overview:

- Hawk-Eye to handover VAR Output in 1080p59.94 to Riedel (x6)
- Riedel to deliver SDI feeds with correct audio to Referee Analysis Room
- Hawk-Eye to provide 3x Quad split feed to TV
- Hawk-Eye to provide 3x PGM dirty feed
- Riedel to provide speakers and headsets with correct audio

HEI Requirements:

- Handover VAR output to Riedel for embedding (per VOR)
- Run 6x 3G feeds to refereeing room and provide router panel

No HEI equipment is installed in this area.



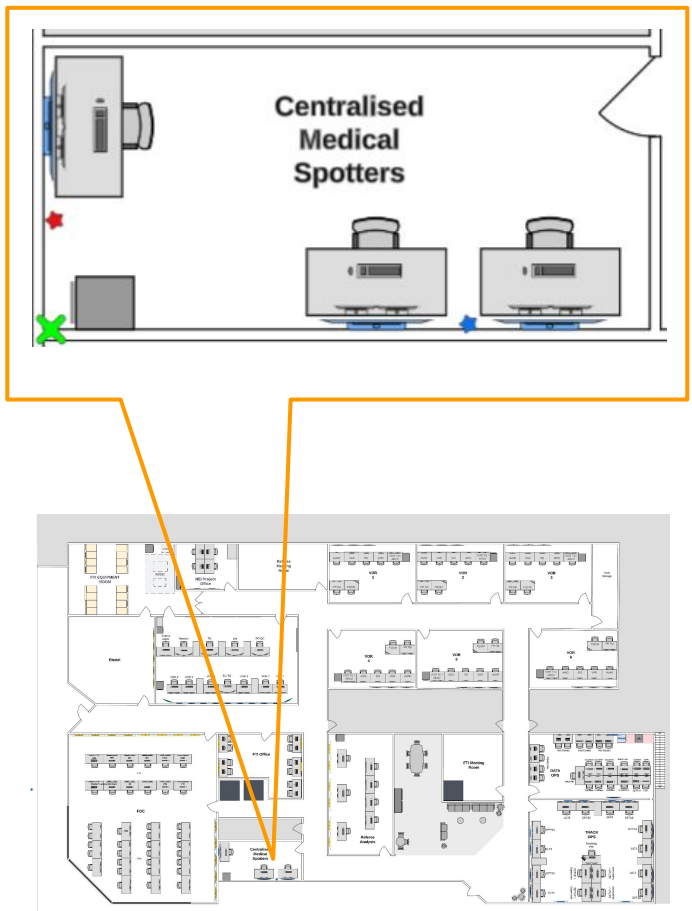
IBC - Centralised Medical

Overview:

- Hawk-Eye to handover 3x PGM in 1080p59.94 to HBS provided and wall mounted TVs). HEI to provide converters
- Provide 3x Medical workstations, (2x Monitors, 1x NUC, Mouse and Keyboard)

HEI Requirements:

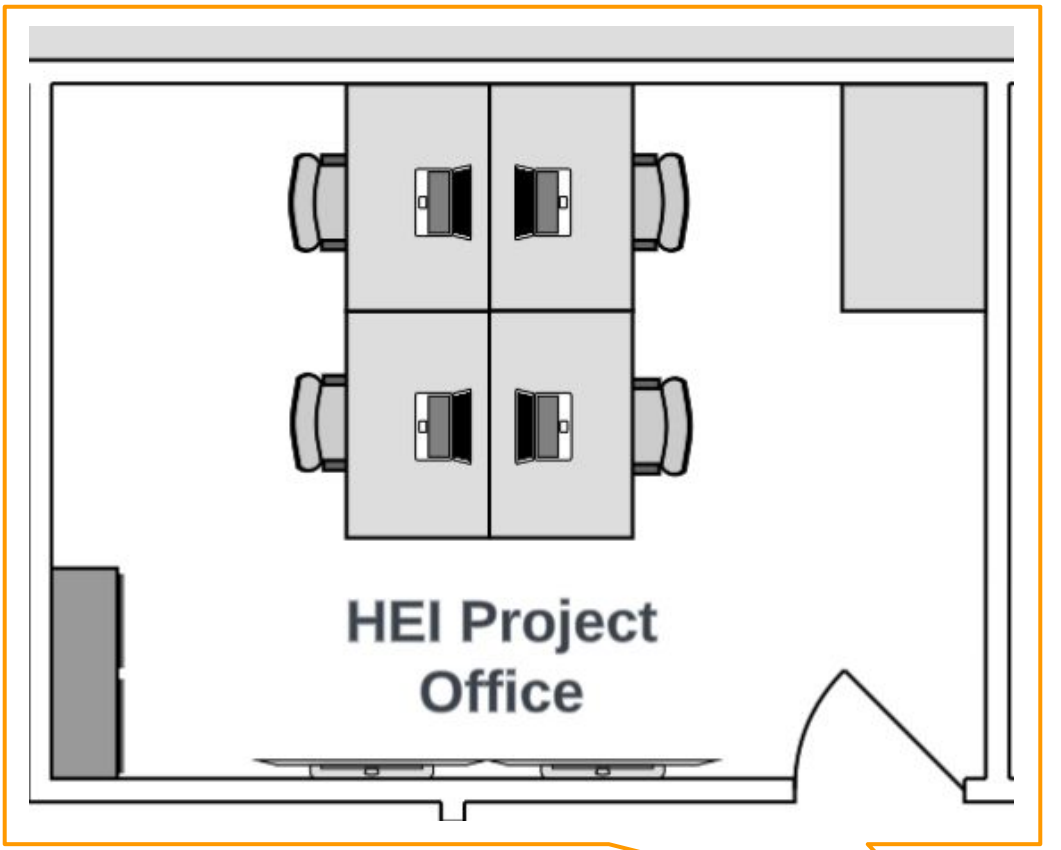
- HEI will provide all signals over fibre
  - Fibre to SDI converters included in kit
  - Fibre SFP+ for switch included
- HEI provide 3x workstations, each with 2x monitors
- 1x 16U rack provisioned
- No audio requirement. Riedel to provide all comms



IBC - HEI Office

- 4x Desks Event Mgmt Team
- 2x HEI supplied, HEI wall mounted, TVs showing PGM

- HEI Project Management Team will be located in here during operational period



GLT / SAOT & KINEXON INSTALLATION & DERIG INFORMATION

GLT, SAOT & Kinexon Installations & Derigs

Who will be responsible for the SAOT/GLT equipment install?

The HEI camera install team. We have 5 teams who will start installations w/c 18th May. The Event Schedule (installation section at the bottom) shows the movements of each team. MTY & GDL are not scheduled because they were installed for FWC26 Playoff Tournament in March.

Who will be responsible for the Kinexon install?

The HEI camera install team. This work will take place at the same time as the HEI SAOT/GLT camera installs. 16 small antennas, 2-4 network switch enclosures & pre-terminated data cabling will be installed. Each venue will have a member from the HEI camera installation team onsite from MD-7 until the opening matchday to assist with HEI & Kinexon set-up. They will also support the installation of the Kinexon equipment at the FTI equipment cabin. After the opening match, the Kinexon system will be setup and stable; any troubleshooting will then be managed by the HEI venue team.

Who will be responsible for the Kinexon & HEI equipment derig?

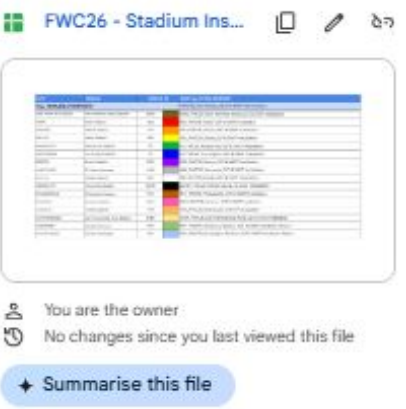
HEI Venue teams. 2 days have been assigned for the removal of 14x GLT, 16x SAOT cameras, 16x Kinexon antennas, network switch enclosures & their data cabling between plus the equipment at the TV compound.

Both HEI & KNX systems are lightweight and being installed using techniques that will be make our derig activities most efficient. Cameras & Antennas will be installed using clamps, magnets & ratchet straps; the use of unistrut will be minimal. Cabinet enclosures will be dribox (Kinexon) & gator cases (HEI). These enclosures will be ratchet strapped (where required) or placed on the floor.

GLT, SAOT & Kinexon Installations & Derigs

Where can I find information about how the SAOT/GLT/Kinexon equipment will be installed at each venue?

At the FWC26 - Stadium Installation Overviews sheet (linked on the FWC26 centralised sheet) you will find the following information:



TABS

- **Reports - GLT/SAOT:** An overview showing the GLT/SAOT schematics, equipment & cabling locations at each venue
- **Reports - KNX:** An overview showing the Kinexon schematics, equipment & cabling locations at each venue
- **HEI/KNX - Fibre:** Fibre core plug up for HEI & KNX between FTI Equipment Cabin at the TV compound & all stadium locations (RRA, BRRA, Media Tribune, Back of Stands / Catwalk)
- **HEI - Network**: Network switch port plug up for the HEI cameras
- **HEI - POE Splitters (Attached to ZCAMs):** An overview detailing which HEI cameras have 171, 172 or Linovision POE splitters installed
- **HEI - POE Repeaters (Cat6 extenders used every 90m):** An overview detailing which HEI cameras have repeaters
- **HEI - GLT Calib (Theodolite Values):** The theodolite measurements for the GLT calibration per venue
- **HEI - Method Statements:** HEI method statements for GLT/SAOT camera installation, maintenance & derig activities
- **Pitch Access:** A high level overview of HEI pitch activities during the camera installation & MD-7 period

GLT, SAOT & Kinexon Installations & Derigs

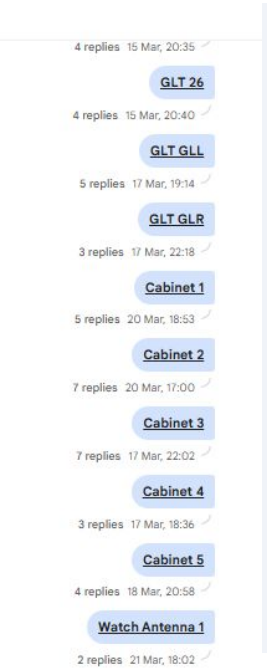
Where can I find information about how the SAOT/GLT/Kinexon equipment will be installed at each venue?

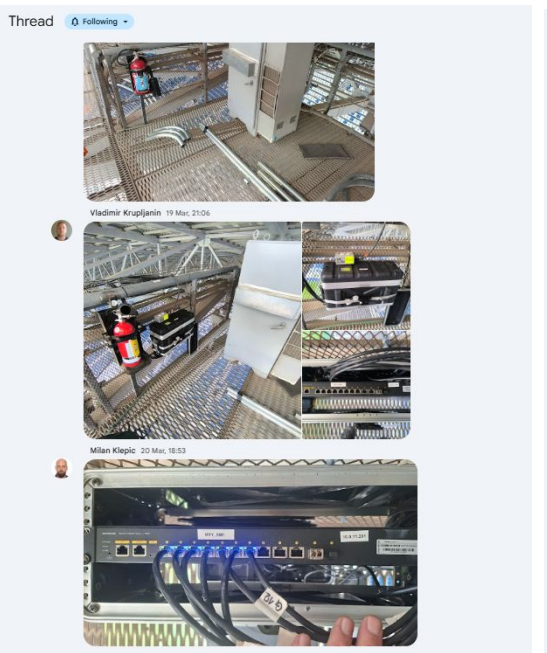
In addition to the FWC26 - Stadium Installation Overviews sheet (linked on the FWC26 centralised sheet) google spaces will be created for each venue where the HEI installation team will upload photos of all HEI & KNX equipment that has been installed.

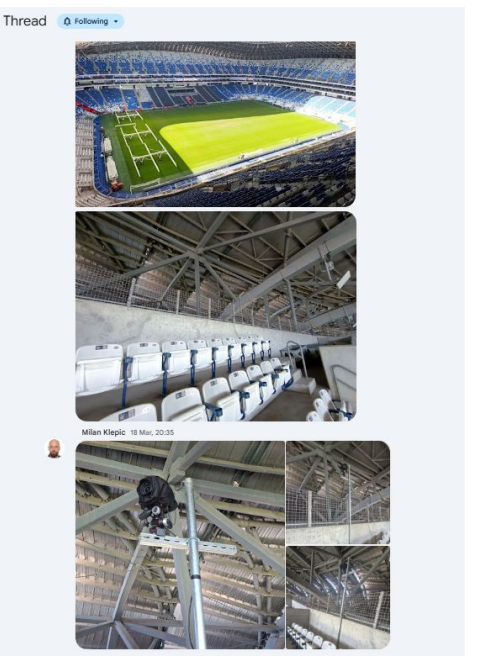
These google spaces will act as a photo library where threads will be created per camera, cabinet & antenna position.

Once the installation period is completed the HEI venue & IBC teams will be added to give visibility.

Installation Google Space - Per Venue







GLT, SAOT & Kinexon Installations & Derigs

Are we using NFL cameras & switches?

Yes, similar to FCWC25, at 10 of the 11 NFL venues in USA, we are planning to use the most accessible HEI NFL equipment for FWC26. However the HEI NFL equipment alone is not enough to cover all FIFA HEI locations, therefore FIFA HEI will supply the remaining equipment.

A high-level breakdown per venue of the HEI NFL equipment planned to be used can be found below;

	HEI NFL EQUIPMENT	
	CAMERAS	NETWORK SWITCHES
ATL	0	1
MIA	26	6
NYNJ	26	5
SEA	20	7
SFBA	8	2
LA	18	0
HOU	18	4
DAL	18	4
KC	6	2
BOS	12	3

ID Stickers: At these 10 venues where there will be a mixture of FIFA HEI & HEI NFL equipment, stickers will be used to help identify the HEI owner.

This will then help when it comes to identifying equipment during the derig and for managing hardware quirks.

HEI NFL Owned

HEI FIFA Owned

The HEI NFL camera modules, lenses & network switches will be removed from NFL locations during the camera installation period and installed at the FIFA HEI locations.

During the venue derigs, at these 10 NFL venues, the HEI venue team will be required to return & restore the HEI NFL equipment to it's original NFL positions, either inside NFL camera housings and/or cabinet enclosures.

A detailed plan for install (pre-event) & restore (post-event) will be made available with the required support put in place closer to the time of these scheduled activities.

Are we using NFL housings?

No.

The NFL camera housings were difficult to work with at FCWC25. For FWC26 all cameras will be installed on standard PTZ heads and inside waterproof covers.

GLT, SAOT & Kinexon Installations & Derigs

Who will be responsible for fibre & power installation at each venue?

HEI is not responsible for this. Neither will HEI be installing any fibre extensions.

FIFA's chosen providers for fibre & power will be responsible for this installation at the locations required. It is expected that this will be completed by MD-7.

An overview of the fibre locations & core plug up per venue can be found on the "HEI/KNX - Fibre" tab on FWC26 - Stadium Installation Overviews.

Similar to FCWC25, are we planning to use POE injectors to prevent cameras dropping out upon their plug up to stadium side network switches?

POE injectors have been packed within the camera installation pallets however we do not forecast using them. They will only be installed if this issue is experienced onsite during the setup.

We are confident that we have identified the issue being related to the POE splitter. For FWC26 we have already planned the stadium side switch plug ups and have balanced the connected cameras to have a mixture of the new and old models. This has already been tested successfully at FIFA events since FCWC25.

An overview of the POE splitter model per camera can be found at the "HEI-Splitters" tab on FWC26 - Stadium Installation Overviews. If you do experience the camera drop out issue please report this to the IBC Tracking Supervisors.

What is the plan for GLT Calibration with Theodolites?

We have prepared 5 units. These units will be of the model we are most familiar with: Leica FlexLine Total Station (TS06)

2 units are HEI owned and will travel out with the HEI camera installation team. The other 3 units have been hired in the UK and will be packed within the camera installation pallet cages at 3 venues.

The 5 HEI camera installation teams will travel to each of the installs with a theodolite unit each and will take measurements on the final day of the camera installation period. Unless there is an issue with the measurements we are not planning any theodolite activities during the MD-7 period.

Theodolite measurements can be found at FWC26 - Stadium Installation Overviews

KINEXON
SYSTEM SET-UP

KINEXON SYSTEM SET-UP

At venue, Hawk-Eye will be responsible for the initial Kinexon system set-up supported remotely by Kinexon technical staff at IBC Dallas.

At each venue, the Hawk-Eye “Install Support” role will manage these activities during the MD-7 period under the supervision of the HEI Venue Manager. After the opening match at each venue, the Hawk-Eye “Install Support” role will depart and it’s expected that HEI involvement with the Kinexon system for the remaining matches will be minimal once the system is up and running.

VENUE ACTIVITIES DURING THE HEI CAMERA INSTALLATION PERIOD

1. Installation of 16 antennas, pre-terminated data cabling & network switches. Kinexon installation reports, network and fibre plug up can be found **HERE**

VENUE ACTIVITIES DURING MD-7 PERIOD - FULL SET-UP INSTRUCTIONS GUIDE

2. Installation of Kinexon small rack within the FTI Equipment Cabin
    - Power
    - Internet
    - Fibre connectivity to stadium side switches
    - Integration with the Hawk-Eye system

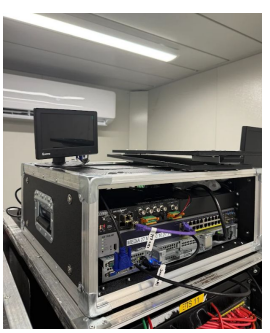
3. Installation of Kinexon’s PTP antenna outside (clear line of sight with the sky) at the FTI Equipment Cabin in the TV Compound. All venues attached to the cabin except MIA & ATL - **GUIDE HERE**

4. Troubleshooting antenna connection issues

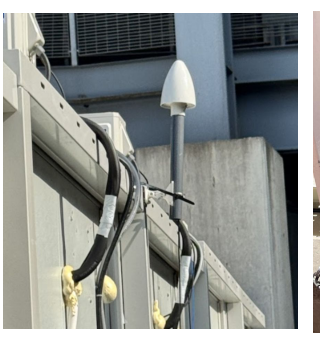
5. System calibration and acceptance testing (approx MD-4) - **GUIDE HERE**
    - Theodolite is no longer required to calibrate the Kinexon system

BALL CHARGING & MANAGEMENT OF THE CONNECTED BALLS

Hawk-Eye are not responsible for charging or the general management of the connected balls onsite, the FTI VM will hold this responsibility. The FTI VM will ask Hawk-Eye Venue Teams to handover the ball chargers from the Kinexon flight case during the set-up period.



Kinexon Base End Equipment
FTI Equipment Cabin

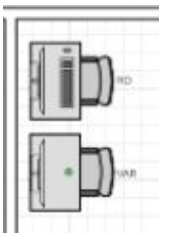


1x PTP Antenna
(TV Compound)

16x Pitch Facing
Antennas

VAR

VAR - Onsite



Setup:
- Backup RO in Ops Room (only applicable for KO's - TBC)
- Backup VAR station in Ops Room (all matches)

Feeds/Routing:
- 5 ISO feeds + PGM will be captured for the backup VAR station
- Reference will be taken
- PGM and Cam 1 will be captured and displayed on monitors.
- There is no SDI router. PGM and Cam 1 will duplicated via a DA
- 16 way multiview feed for monitoring purposes will be streamed back from IBC
- 4 way multiview will be generated in the rack which will be streamed back to IBC for monitoring
- There will be a decoder in the rack and a backup stream for RRA feeds
- HBS will provide a small SDI gateway adjacent to the rack in order to receive the feeds. This is the handover point
- We will be providing clean goal line cameras to HBS at venue

Note: All HB inputs and outputs will be HDR

RRA:
RRA Main - Pyramid
(Power & Fibre provided by FIFA, shared with Reidel, fibre patch lead provided by us)

RRA Backup - Tabletop
(Power & Fibre provided by FIFA, used for Medical & Analyst tablets also)

Privacy graphic shall be tournament logo, set by production and streamed from IBC.
Take note of the difference between RRA privacy, HEO test card and IBC test card.

Ensure RRAs are clean, tidy and presentable. There should be no other logos / branding / labels etc visible on the RRA(eg from previous competitions). FIFA are likely to brand the Pyramid RRA in tournament colours / graphics.

Note the Main and Backup RRAs always have the same privacy output. Unlike some domestic leagues, the backup RRA shall not show an output from another viewer

RRA Main may need to be moved into position after team walkout. We have informed FIFA that we do not have staffing onsite available for this under the new delivery model.

Audio Ref Comms:
Audio - Riedel will be predominantly located in the IBC. They will have some equipment / representatives onsite. Equipment is likely to go within the FTI Equipment Room.

Riedel should install various panels and headsets into the Ops room for the backup VAR station

REPLAY DEVICES

Overview

We will be setting up 5x devices pitchside - 2x devices per team bench/technical area and 1x device for FIFA Medical (BRRA position).

Check venue specific plan for Media Tribune positions. If in doubt, FTI Manager will confirm all locations.

Each venue has unterminated cat6 along with cat6 heads/boots to be run and crimped to the length we need from CDP switch to device. *This will be completed by the HEI installation team.*

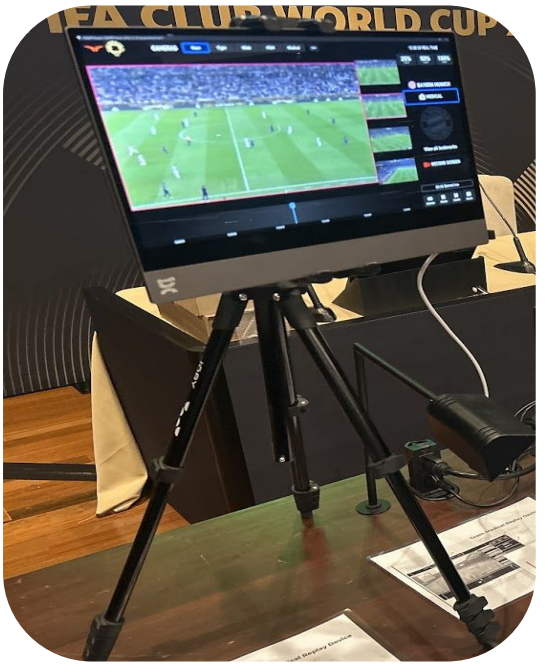
The setup consists of:

- Small form factor NUC in a waterproof peli box
    - Pitchside: 15” slim Espresso touchscreen monitor mounted on a tripod
    - Media Tribune: 17” slim Espresso touchscreen monitor mounted on a desk stand
- Cabled network to the NUC
- Long 4m USB-C cable from the NUC to a 15” Espresso monitor
- Shorter purple/white 1m USB-C Cable from the NUC to a 17” Espresso monitor
- The use of SMARTRemote to connect back to servers running SMARTViewer in the IBC

In this setup, the Peli Case containing the NUC where SMARTRemote is running can be positioned away from the screen, e.g. under the seats and reduces the risk of the network cable becoming removed.

Please take care with the screens to ensure they are not dropped, scratched or smashed!

Do not refer to positions as Home and Away, they should be Team A and Team B



Replay Devices - Pitch Side

* Photos not fully representative of equipment to be used

FIFA Medical - Pitchside x1

- Maximum quad split
- Backup RRA position
- 15” Espresso touchscreen monitor with tripod and tablet holder
- NUC/SFF in Pelicase and placed out of view
- 4m USB-C cable from NUC to Espresso monitor.
- Local power (from LOC/FIFA)
- CDP switch at BRRA location - cat6 cable to NUC in pelicase

Team Devices - Pitchside x4

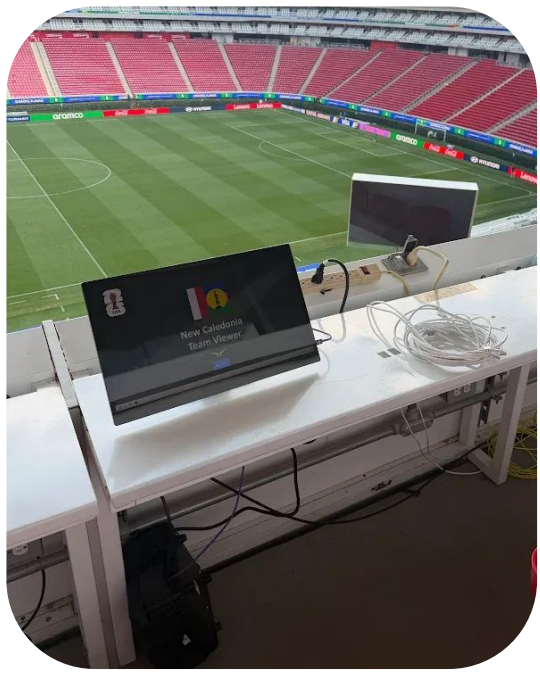
- 2x per team bench / technical area
- 15” Espresso touchscreen monitor with tripod and tablet holder
- NUC/SFF in Pelicase and placed out of view
- 1m USB-C cable from NUC to Espresso monitor.
- Local power (from LOC/FIFA)
- CDP switch at BRRA location - run cat6 cable to NUC in pelicase

Replay Devices - Media Tribune

* Photos not representative of GUI / Layouts to be used

2x Team Analyst Viewer - Media Tribune

- Team A and Team B positions
  - NOT HOME AND AWAY!
- 17” Espresso touchscreen monitor, with KB and mouse
- Place NUC/SFF in Pelicase and placed out of view
- Cat6 from CDP switch located centrally at Media Tribune
- Local power (from LOC/FIFA)



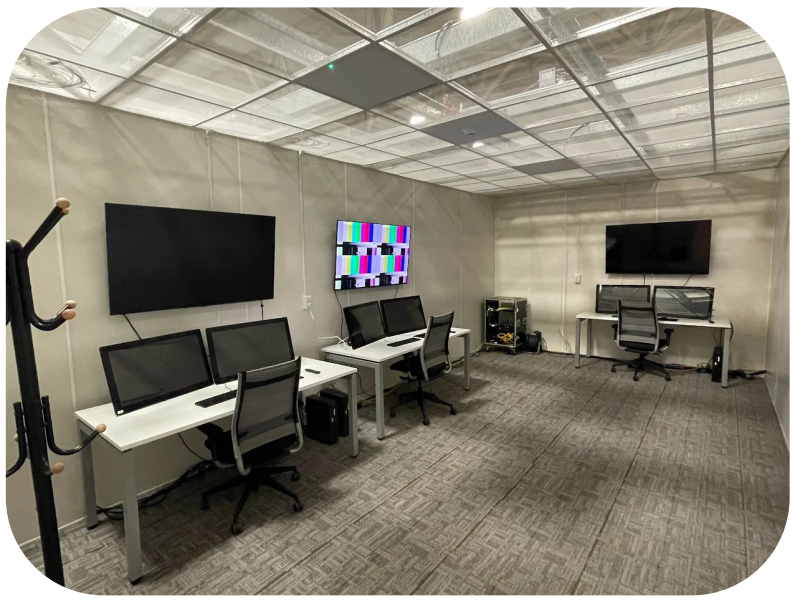
FIFA Medical Replay Device - IBC

The FIFA Remote Injury Spotters are located in the IBC for this event.

1x Z2 per station

These stations will be remotely connection to the venue by HEI technical staff at IBC

All comms are managed by Riedel

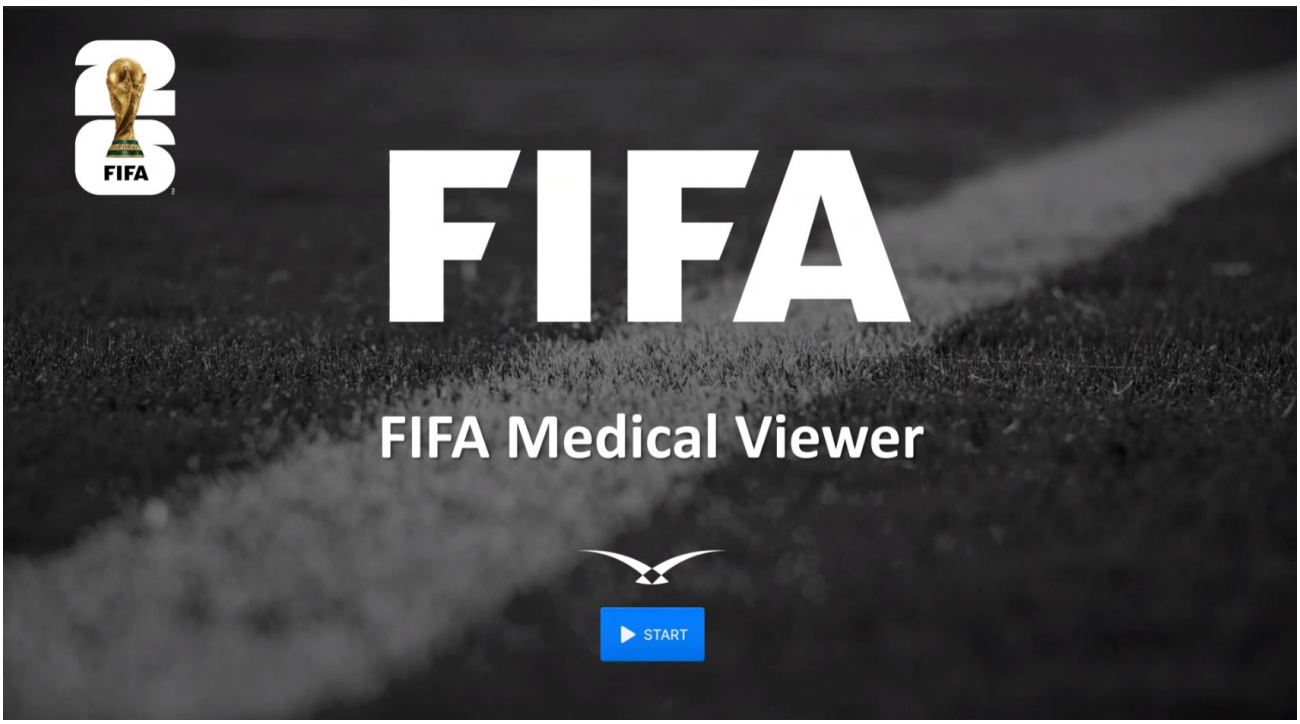


Replay Devices - GUI

A new GUI / Layout has been created and agreed with FIFA for the various devices / roles. This should all be deployed by configurator via productions.

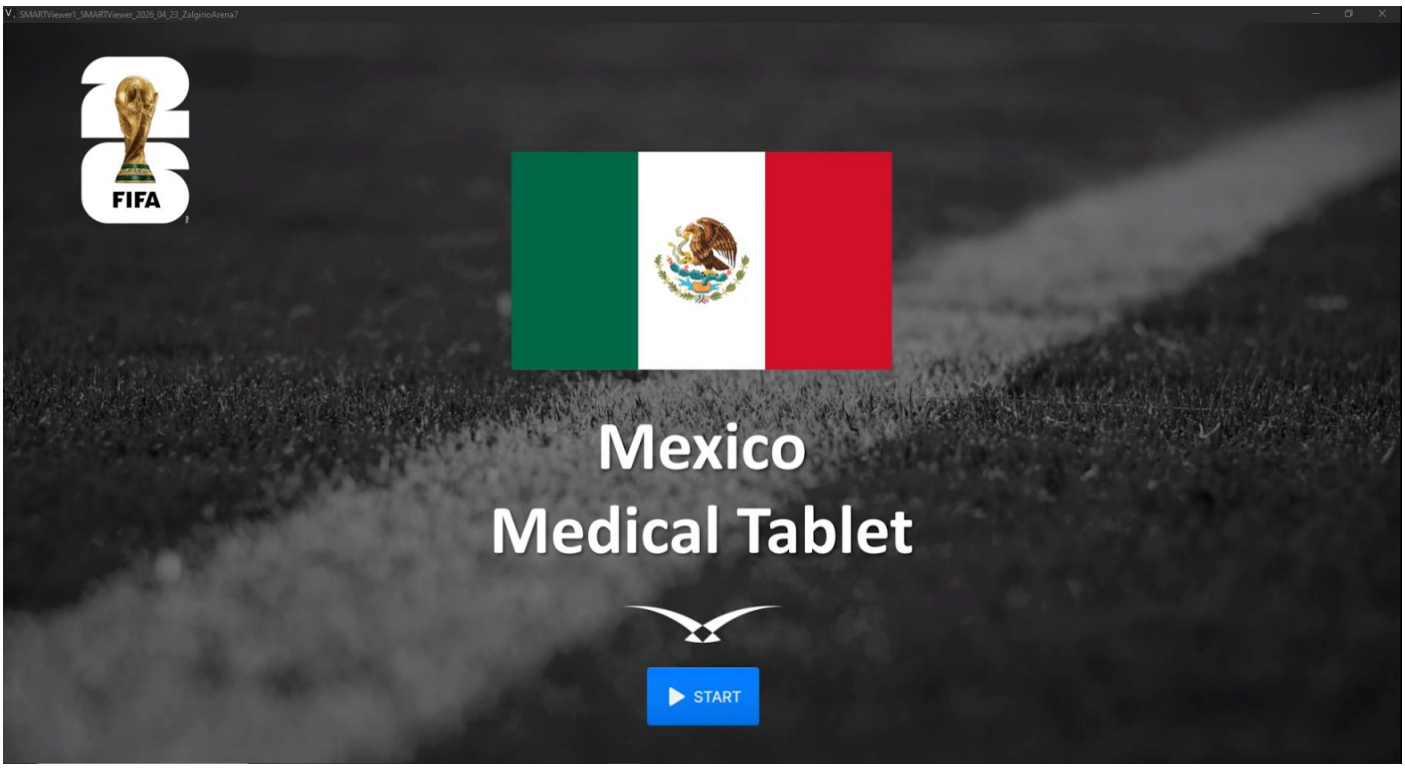
FIFA

FIFA Medical Viewer (IBC) & Tablet (Pitchside at Venue)



Replay Devices - GUI

Team A



Replay Devices - GUI

Team B

GLT

Overview

Hawk-Eye will provide Goal Line Technology at all 16x stadiums of the FIFA World Cup 2026. The service delivery will be inline with FCWC25 whereby core GLT operations take place from the IBC.

The 1st ball has been confirmed for all matches except for Semi-Finals onwards. Adidas will be releasing a 2nd ball ahead of or during the tournament for Semi-Finals and Finals.

Key Points:

- 12x tracking cameras (Zcam)
- 2x Goal Line Ultramotion (ZCam)
- Network switch topology
- Box Left, Box Right feeds to TV handed over **onsite** (p59.94 10-bit 3G HDR)
  - Box Left = Goal Line Left,
  - Box RIght = Goal Line Right
- VR feed is new UE5 VR, VR Op operates in VOR at the IBC (Handed over at IBC)
- GOAL audio to the match officials
- 8x Silver Leikr watches, same as used for FCWC25

Ball (Trionda):

- 2 ball versions will be used at the tournament
  - White for Group Stage / R16 / QF
  - TBA for Semi Finals & Final
- Due to Kinexon integration, the match balls will have a “chip” in it
  - Essential that this match ball is used for all match-day pitch activities including sync

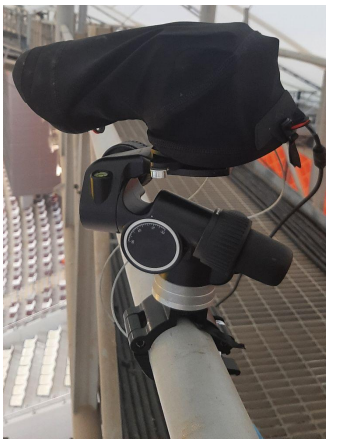
Ball 1
All Group Stage / R16 / QF

Ball 2 - TBA
Semi-Final, 3/4 Playoff & Final

System Design

Based on the actual distribution of cameras / cabinets, :

- 1x 10Gb link from each base end switch to associated tracking cabinet (switches)
- GLT and EPTS tracking cameras through the same infrastructure separated by VLANs
- Cabinet switches (Netgear M4250) are PoE - powers all tracking cameras (GLT and EPTS).
    - Ports 1-4 = GLT - Cat6
    - Ports 5-8 = EPTS - Cat6
    - Ports 9,10 = Spare - Cat6
    - Port 11 - Link to base switch - 10Gb fibre SFP+
    - Port 12 - Spare Link to base switch - 10Gb fibre SFP+
- All GLT cameras will have set IPs
    - 10.0.11.11 - 10.0.11.16 - End 1
    - 10.0.11.21 - 10.0.11.26 - End 2
    - 10.0.11.41, 10.0.11.42 - UM1, UM2
    - 10.0.2.31,10.0.2.32 - Watch1, Watch2 **(Note watches are on .2 network)**
- Watch transmitters on the catwalk as normal. This will be connected to a POE fibre-Cat6 converter and connect directly to the .2 network switch
- UM1 and UM2 will be Zcam. Cat6 direct to module, powered by PoE
- Housings will all be soft covers (when using NFL cameras / positions cams will be removed from hard cameras enclosures)



All IPs can be found on the IP Usage sheet

GLT VR Workflow NEW

As noted previously, there is a change in typical GLT VR workflow. The main change is that a dedicated VR Operator, located in the VOR, will be generating both VR (clean, and GDS replays) as well as AOT VR replays.

Viewer/HawkVIS software shall be used which takes in the necessary feeds:
Goal Line, EPTS and Broadcast cameras

The SDI output of this should be handed over to TV (HBS at IBC) as the “VR Replay” feed

Clean replays can be offered in the viewer by playing out GLL or GLR in Viewer (F1 and F2 as normal)

GLT VR replays (GDS) should **only** be used when a close GDS replay needs to be generated

UM Colours

As this stage, it is expected that colour balancing the goal-line UM cameras will be completed by the GLT Operator. Colour balancing will be done via the Z-Cam WebUI and need to be accessed from a venue server.

The GLT operator can speak to colours by using the talkback panel

Staffing

Each match will have dedicated GLT Operator (IBC) , VR Operator (IBC) and TTG (Onsite)

GLT Operator:
- General responsibility of accuracy and operation of full system, including Calibration, checklists,, and reporting

Onsite TTG / Ad Serv:
- Line-ups, On pitch calibration, Sync test (for VAR), Watches to match officials changing room, On pitch support for referee test

VR Operator
- clean and virtual replays

GLT/EPTS IPs (all ZCams)

Service	Machine	.2 Network	.3 Network	Cameras	KINEXON
All	Server 1	10.0.2.141	10.0.3.141	10.0.11.51 10.0.12.151	10.10.2.211
All	Server 2	10.0.2.142	10.0.3.142	10.0.11.52 10.0.12.152	
All	Server 3	10.0.2.143	10.0.3.143	10.0.11.53 10.0.12.153	10.10.2.213
GLT/VAR	Watch 1 Modem	10.0.2.31			
GLT/VAR	Watch 2 Modem	10.0.2.32			
EPTS	EPTS Camera 1			10.0.12.101	
EPTS	EPTS Camera 2			10.0.12.102	
EPTS	EPTS Camera 3			10.0.12.103	
EPTS	EPTS Camera 4			10.0.12.104	
EPTS	EPTS Camera 5			10.0.12.105	
EPTS	EPTS Camera 6			10.0.12.106	
EPTS	EPTS Camera 7			10.0.12.107	
EPTS	EPTS Camera 8			10.0.12.108	
EPTS	EPTS Camera 9			10.0.12.109	
EPTS	EPTS Camera 10			10.0.12.110	
EPTS	EPTS Camera 11			10.0.12.111	
EPTS	EPTS Camera 12			10.0.12.112	
EPTS	EPTS Camera 13			10.0.12.113	
EPTS	EPTS Camera 14			10.0.12.114	
EPTS	EPTS Camera 15			10.0.12.115	
EPTS	EPTS Camera 16			10.0.12.116	
GLT	GLT Camera 11			10.0.11.11	
GLT	GLT Camera 12			10.0.11.12	
GLT	GLT Camera 13			10.0.11.13	
GLT	GLT Camera 14			10.0.11.14	
GLT	GLT Camera 15			10.0.11.15	
GLT	GLT Camera 16			10.0.11.16	
GLT	GLT Camera 21			10.0.11.21	
GLT	GLT Camera 22			10.0.11.22	
GLT	GLT Camera 23			10.0.11.23	
GLT	GLT Camera 24			10.0.11.24	
GLT	GLT Camera 25			10.0.11.25	
GLT	GLT Camera 26			10.0.11.26	
GLT	UM1 (Z Cam)			10.0.11.41	
GLT	UM2 (Z Cam)			10.0.11.42	

Feeds

Given we are providing VAR, GLT and EPTS, HB will only hand over feeds once for HEI to distribute out internally between services.

You should receive the following feeds from the VAR TG:

- Reference (tri-level or B+B)
- Feeds for colour balancing can be pulled across the network for CCU for scopes:
  - Main Camera 1
  - 16m Left
  - 16m Right

You should provide the following feeds to the VTG to pass onto TV in the correct production format

- Goal Line Left, (Venue)
- Goal Line Right, (Venue)
- VR (IBC)

All feeds to TV must be 1080p59.94 HDR (HLG)

GDS Replays NEW

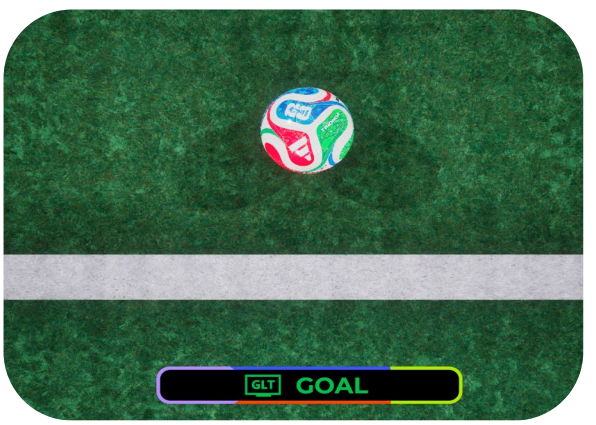
The GLT VR Replay look and feel will remain in line with previous tournaments but with **updated UE5 graphics.**

All VR ops must practice GLT VR replays in order to provide a high quality GDS replay to TV within 10 seconds of the incident occurring, taking note of the new workflow

This will be monitored and recorded by the QCMs.

The graphics can be seen below for both balls:

Non-Finals Ball



Finals Ball

The finals ball graphic has been developed but is not available to share due to the ball not being released. Once the ball has been released we will share it with the operations team

Field of View and Replays

Given the goal line cameras will be used for EPTS, the field of view / camera lineup must include the full penalty box for this and VAR purposes, but be good enough to use for GDS replays.

You will be requested to send a grab of your field of view / camera lineups to IBC QCR team for quality control purposes. The field of view / lineup must be unobstructed, have consistent focus and be vertically parallel to the goal line

When providing GLT VR (i.e. GDS) replays, a cutout will be used in VR to ensure a field of view closer to the goal line

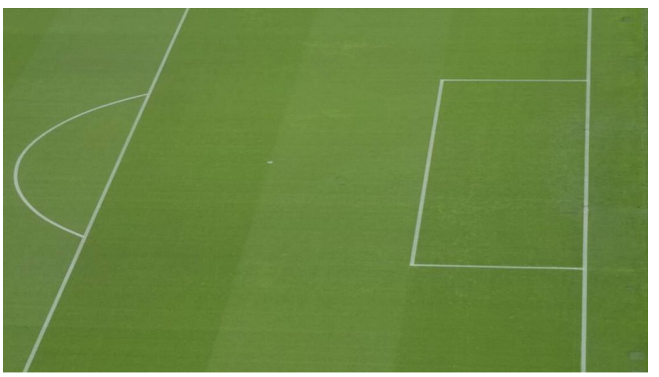
When providing clean replays, the GLT VR operator has the flexibility to provide it either with the full field of view or to zoom in on the viewer, depending on the type of incident / replay

Replays will typically go via the coordinating producer/VT/EVS

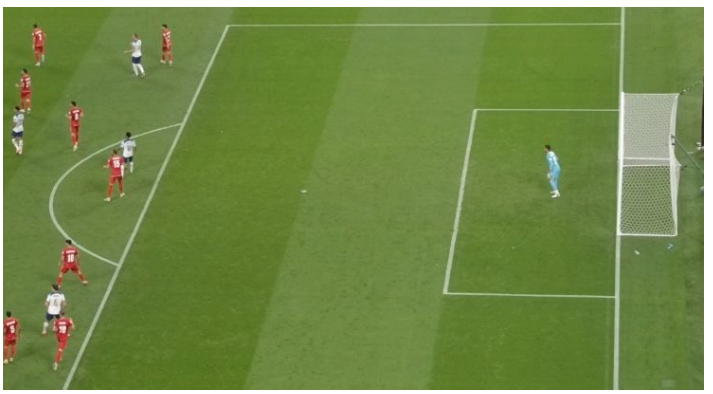
All VR Ops must be familiar with this prior to the tournament and be able to quickly push out replays of interest to the HB

UM full image - can be used for clean replays

VR - only to be used for GDS animation



Zoomed in UM - can be adjusted for each stadium on viewer to provide replays to TV that are more zoomed in then the full image



Leikr Watches NEW

8x Leikr watches will be provided to the match officials / FIFA for each match. Typically this is allocated:
In the Box:

- 4x for the on field team
- 4x spare for reserve referee etc

4 of the watches will be provided with a **velcro strap**, whilst the other 4 will be standard rubber strap. The straps can be moved between watches as required. Please seek further advice on how to do this to avoid damaging / breaking the watch or strap

It is the responsibility of the VM to ensure these watches are handed over at the correct time, fully charged and ready for use. The box should remain plugged in and charging when not in use / in the changing room pre-match

The watches and chargers will all be built into a box (similar to UEFA) which should be taken to the match officials changing room and plugged into power.

Notes:

- It is essential the referee/AR1/AR2/4th are allocated the correct watch - the role will be shown on the screen
- Please reset the log prior to each match. If in doubt, seek further advice
- **To guarantee maximum battery life, it is standard operating procedure to mute GOAL alerts on control during the warmup. The constant spamming of triggers to the watches will quickly drain their battery** NEW

AOT / EPTS

Overview

Hawk-Eye will provide its AOT System at all 16 stadiums of the FIFA World Cup USA, Canada & Mexico 2026.

- 16x tracking cameras (ZCam) will be installed into each venue.
    - 4 cameras will be available for AOT replays and these will be colour balanced to a broadcast standard. These cameras will be 1, 8, 9 & 10.
- Integration of 4 broadcast offside cameras for tracking & AOT Replays
- Audio alerts will be played to the onfield refereeing team for offside decisions.
- PTP sync from Kinexon.
    - Kinexon are responsible for providing it to the FTI Equipment Room
- Live & Delayed Tracking Data sent to FIFA, Kinexon, and other consumers via normal internet delivery methods
- Automatic Offside system for Officiating & VAR use.
- VR replays of offside decisions to HB.
    - Operated at IBC in the VOR, first replay for automatic playout will be in use as well as replays on broadcast cameras.
    - Goalkeeper Line of Sight to be used via legacy VR
- Integration with Kinexon kick points.

Staffing

The following roles have been assigned for the tournament:

- 1x EPTS Operator per match (5x in total) - IBC
- 1x Tracking Supervisor (GLT & EPTS) per match (4x in total) - IBC
- 1x Data Supervisor per match (2x in total) - IBC
- 1x VR Operator per match (3x in total) - IBC VOR
- 1x Tracking Guarantee per venue (16x in total) - Venue

System Design

This AOT hardware was specifically procured and configured for FIFA tournaments.

Kits have undergone an update before the start of FWC26. Updates include:

- ***Higher performance servers to take account of additional cameras.***
- ***Adjustments to networking given GLT, VAR and EPTS servers will all be together.***
- ***Removal of some of the patch panels to streamline installation onsite.***

Additionally, all GLT cameras & EPTS cameras will be on the same infrastructure, separated by VLANs in the switch configuration.

Feeds

Given we are providing VAR, GLT and AOT, HB will only hand over feeds once for HEI to distribute out internally between services.

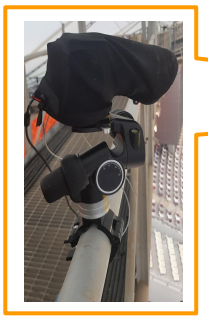
VR Operator, located in the VOR, will be generating both GLT VR (clean, and GDS replays) as well as AOT VR replays. Therefore we will not hand over a dedicated AOT VR feed

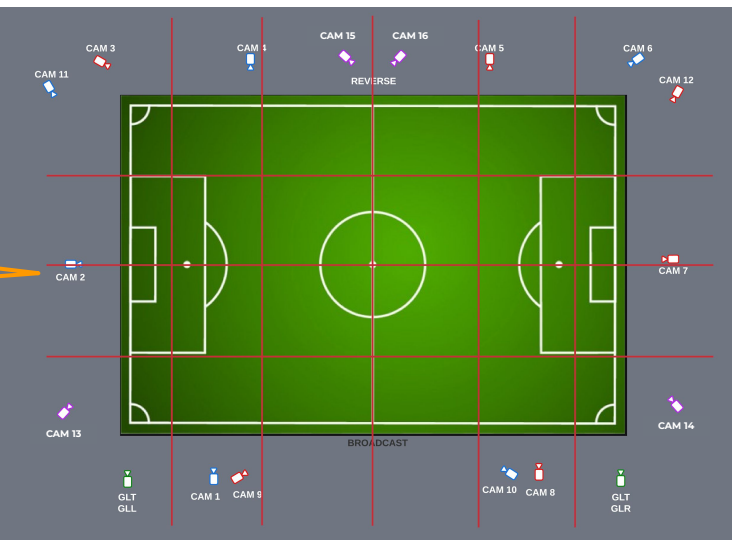
Schematics

There will be 6 extra cameras used at FWC26 compared to UEFA. These will be cameras 11, 12, 13, 14, 15 & 16.

The schematic below is generic and shows the set-up when all cameras are on the same level.

LINE UP GUIDE





Kinexon - Integration

Kinexon provide the system which uses the "chip in the ball" to generate ball telemetry data. This data is used in our system to help match up kickpoints with our (S)AO system. It also allows the VAR team to review potential small deflections of the ball when adjudicating possible handballs, offside touches etc

Kinexon will provide us with both the data feed and PTP sync signal. These will be handed over to HEI as a switch and fibre, going into the EPTS Base End 1 1G port. The Kinexon Data Receiver Application will run on Track 1 with the master TSS & Time Client. **Camera Sync Status (CSS) is now our monitoring tool for camera syncs.**

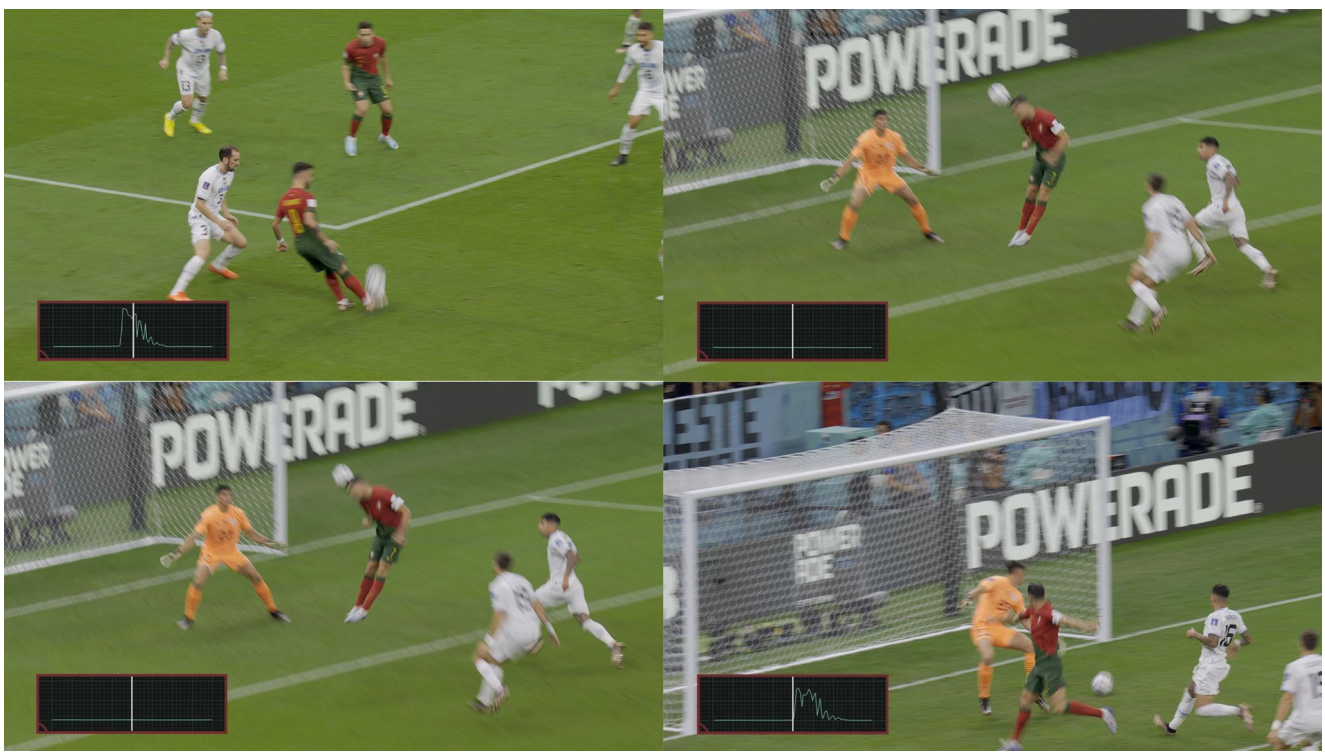
Kinexon will also be subscribing to our RMQ queue.

The TTG will be the main point of contact with Kinexon at each venue.

The HEI venue team shall set-up the Kinexon equipment on their behalf. This will include turning on their server, plugging up cables and remotely troubleshooting NEW

Kinexon Testing

There will be a testing schedule with Kinexon during the set-up period and also each match day. This schedule will be communicated in advance to those who are involved.



AOT Alerts

AOT Alerts will be sent to the onfield refereeing team as seen since the Olympics, with variations for Video Support tournaments. This takes into account both the distance of the offside decision, and the confidence that the system has in the decision. Significant amounts of data will be collected from these matches to help improve the AOT system in future tournaments.

Tolerance Levels

- Clear Offside: **>0.1m**
- Close Offside: **0.1m - 0m**
- Close Onside: **0m - 1.5m**

Audio Alerts

EVENT	CONFIDENCE	ICON	AUDIO	WATCH MESSAGE	ON PITCH TESTING (4 people)
Clear Offside	High Confidence		OFFSIDE (x2)	No	>0.1m Distance
Clear Offside	Low Confidence		DELAY (x2)	No	>0.1m Distance
Close Offside	High Confidence		TIGHT (x2)	No	<0.1m Distance
Close Offside	Low Confidence		DELAY (x2)	No	<0.1m Distance Add 5th number to line up

Data & Deliverables

Live & Delayed

Live & delayed feeds will be sent to multiple consumers for each match. A breakdown of these is as follows:

- FIFA - Live FIFA Uniform
  Delayed FIFA Enriched
  Delayed FIFA Enriched Joints
  Delayed Summaries (Segment, Match-Minute, Match Summary)
  FIFA Events (Auto, Manual, & Combined)

- BeyondSports - Delayed FIFA Enriched Joints
  FIFA Events

- Kinexon - Live Ball (legacy)
  Live Centroids (legacy)

Scrubbed

Scrubbed tracking data (enriched, enriched joints, & summaries) will be sent to FIFA post-match, with a 1hr deadline for upload.

AOT Replays

SAOOD will be in use for this event. As a result we will be running SAO-OD (SAO replays over broadcast cameras 16mL and 16mR) as a live deliverable .

Cameras 1 & 8 will be aligned to get the full width of the pitch in the field of view. Where possible they will also get the full half of the pitch (length). Where this cannot be achieved, other cameras will be used to cover any blind spots from a backup replay point of view.

1st Replay prioritized broadcast cameras where possible and fails over to

1st Replay Doc

You will be requested to send a grab of your field of view / camera lineups to IBC QC team for quality control purposes. The field of view / lineup must be unobstructed, have consistent focus and be horizontally parallel to the near touch line.

FWC22 SAO Replay Cameras 1 & 8 Examples

VR Operations NEW

All VR operations (GLT and SAO) are now conducted by an operator in the VOR. They will be working closely with FIFA and on comms to GLT op, EPTS op, RO, ARO, ARO2 and FQCM in order to provide the following:

- Full frame rate replays from goal line left and goal line right
- “GDS” decisions
- AOT replays (on broadcast cameras)

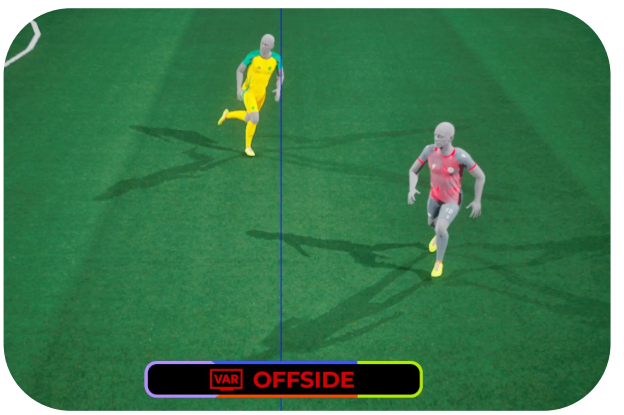
The operator will control what gets sent to TV using a viewer running on VR machine. This Viewer will also do the necessary SDR-HDR upscale

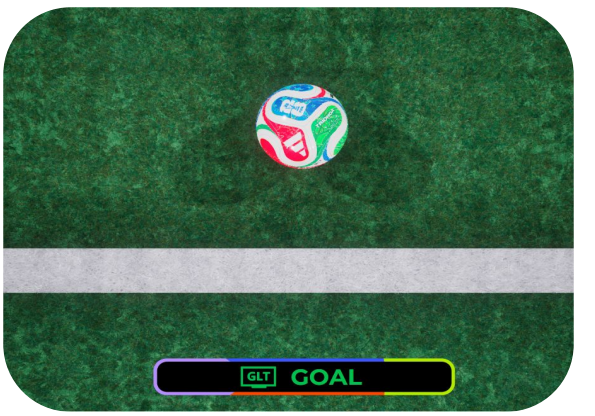
Getting replays played out to TV quickly but accurately is vital

e.g. following an offside decision or close onside leading to a goal.

Colour balancing of our cameras will be completed by another operator. Therefore the VR operator shall concentrate entirely on building and playing out replays

Goalkeeper line of sight (for subjective offsides / interference) will be used live. It will be the VR Operators responsibility to have the GK LOS VRs setup with the correct map and kits.





EVENT DATA

Overview

In line with recent FIFA tournaments, we will be the official data collector for the competition.

Event Data delivery requires the following per match:

- 1x Event Data Collector (EDC) at each venue
  - The EDC will be responsible for collecting manual events quickly and accurately, and entering them into our event data collection tool (FELA)
  - They will have a laptop running FELA and a monitor for live PGM feed (provided by HEI, streaming the PGM feed over a provided VLAN from the FTI Equipment Room)

- 1x Event Data Reviewer (EDR) & Event Data Speaker (EDS) based at the IBC.
  - The EDR/EDS team work with the EDC to add secondary / additional information to the events. The EDR is able to correct any events if any details are difficult to see in live play.
  - The EDR will have the live PGM and Cam 1 feeds, as well as access to a machine running a review tool which allows them to review past events.
  - The EDR/EDS and EDC are in continuous contact by an audio call, and are also able to listen to the venue match director.

- The Event Data Supervisor is based with the EDRs in the IBC. They will be in contact with FIFA for any data issues or any events which need clarification (e.g. goalscorers, attempts on goal)

EDC - Onsite

EDR - Remote
(image of EDR in ROC from FIFA Futsal World Cup 2025)

REPLAY OPERATIONS

RO

The ROs will be located at the IBC

As with previous tournaments, the ROs may travel onsite from quarter finals onwards as a backup solution in case the connection between the stadium and the IBC get lost.

The ROs will oversee the match preparation for the VAR system. It is their responsibility to complete the checklist accordingly to make sure everything is set and tested before KO.

At the end of the match, the ROs must complete the exports required and it is their responsibility to make sure all the clips are exported correctly, with the right feeds and audio requested.

UPDATE: When checking audio on your clips, please play the clips from Main Viewer and toggle open atmo on your Riedel talkback panel in order to hear the audio from the archive

Also, the ROs will be responsible to complete the match report section assigned to them.

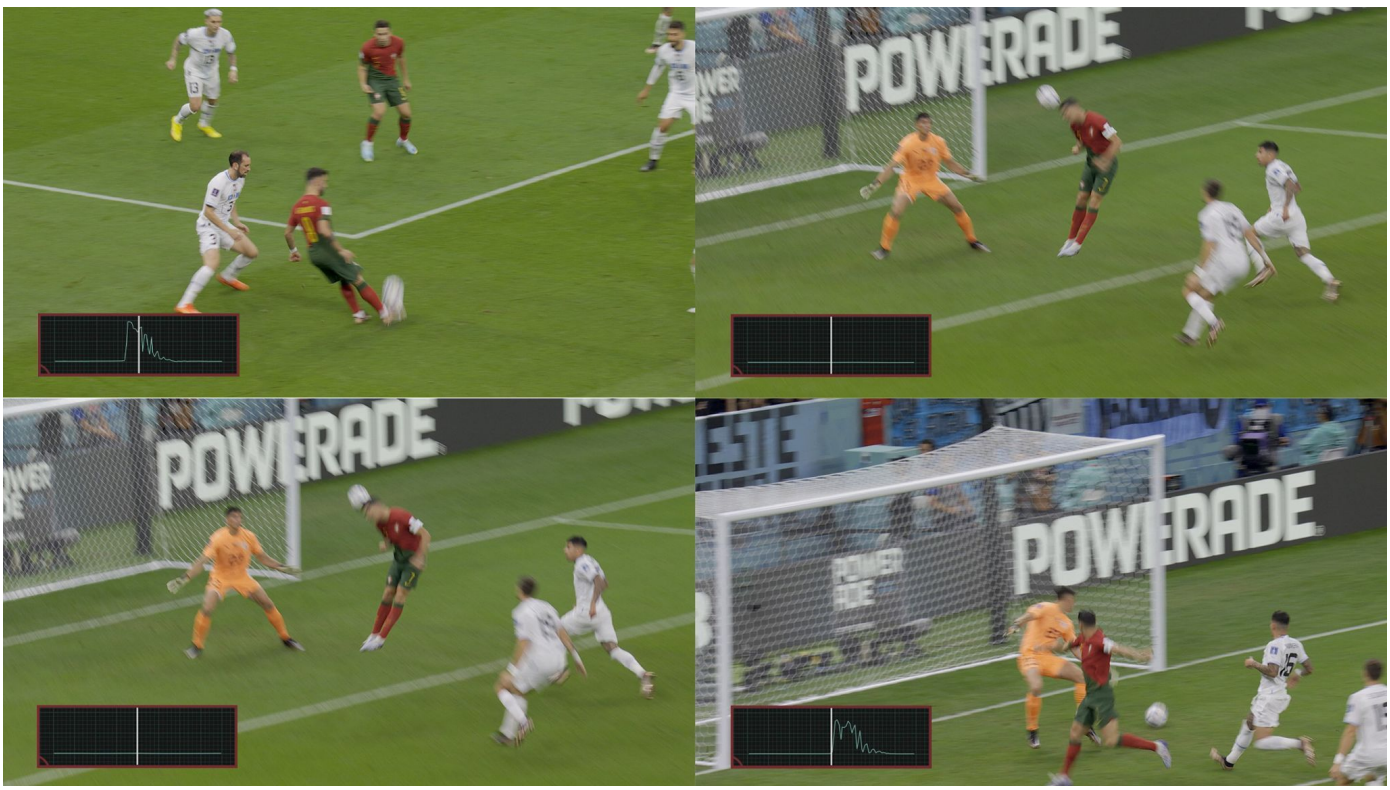
ARO

The AROs will be located at the IBC and responsible for setting up the VAR system in the build-up to the match by working alongside the VOR TG to complete sync, pans, internal testing, checklists etc. Assisting the RO during live operations and relays any issues to the ARO 2/TG.

ARO 2 / TG

The ARO 2 / TGs will be located at the IBC and will join approx 3-4 hours before K/O to perform QA checks. During match operations, the ARO2 will monitor and operate the Kinexon Viewer for ball telemetry data. They will also be the eyes and ears of the VOR TG and will be the point of escalation for any technical issues. The ARO2/TG may be assigned up to 2 matchers per day.

Kinexon telemetry (ball accelerometer) example



RO QC Station

Following the success of the rollout at FCWC25, we will have an RO Quality Control station located in the Hawk-Eye Technical Ops room. This role will be staffed by ROs via the RO lead to those who have not been assigned to matches.

The main aspect of this role is to review our peers and to share knowledge to improve our operations. To do this, Samir will create a checklist to be followed.

Another important aspect of the role is note down any VR replays to air so they can be clipped for FIFA Refereeing.

Finally, the Duty Manage / Technical Manager may refer to this person to gain expertise advice on any incident they may see or if FIFA ask operational questions.

Archives

Archive Guide.

For the avoidance of doubt, it is the responsibility of the main **Replay Operator** (not ARO or ARO2) to ensure all archives are correctly completed for the match. The VTG / VM / ARO etc can assist them, however the RO remains ultimately responsible for this.

"Correctly completed" includes:

- ✓ Correct feed / angle
- ✓ Duration
- ✓ Format
- ✓ File name
- ✓ Audio
- ✓ Copied to the relevant NAS drive at IBC

NAS

A NAS drive will be located at the IBC. All full match archives and refereeing clips should be stored on these NAS's. These shall then be copied to FIFA hard drives as and when required, and synchronised to the S3/AWS bucket we run for FIFA. This will be done by the mgmt team

Other Archives

The RO will also be responsible for collecting other data archives through viewer that are required. Such additional archives could include:

- Offside incidents including metadata, Elastic bookmark, SAOS data
- CVIDs of any training incidents

The checklist will detail any further data required

HBS Datatainment Operations

HBS Datatainment

HBS Datatainment is a live service that will provide 2D & 3D visual and data overlays over a traditional program experience for every match. We achieve this by combining our our player tracking and main camera calibration with data science event data allowing us to trigger graphics on screen or above specific players at appropriate times.

Features of the service include:

- A live pitchmap graphic showing player positions at all times
- A momentum meter showing the balance of play over the last 5 minutes
- Shot distances & speeds, run speeds
- Goal shot placement
- Player possession/under pressure markers

Based in the HBS section within the IBC in Dallas, we'll be providing a multilateral feed for HBS and a single unilateral feed, with different sponsor graphics for DAZN Japan.